

22050 Continuous-Time Signals and Linear Systems

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L10

Quantitative features of step response

Bode plot 1: Constants and linear factors

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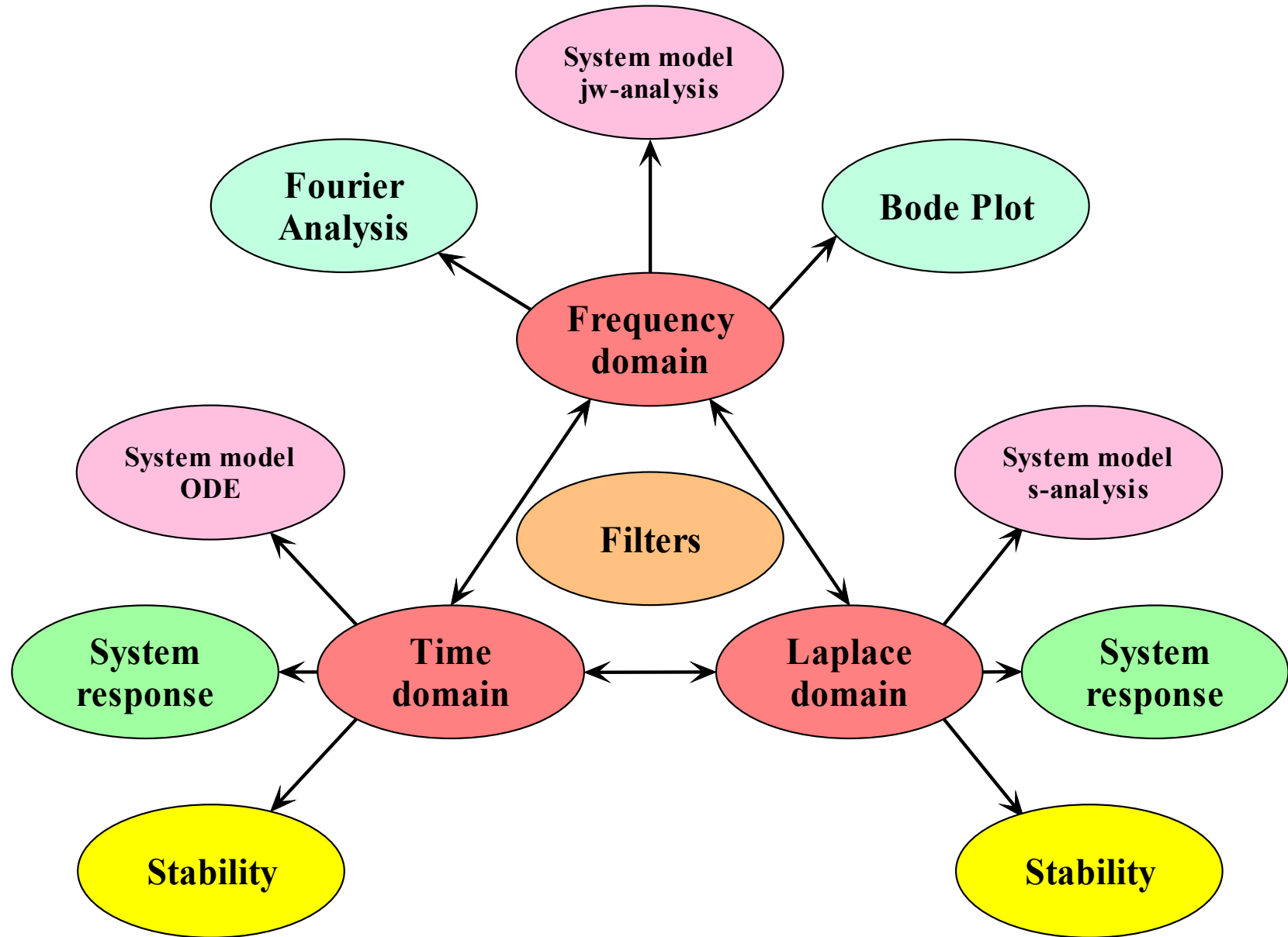
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Analysis of second order system

6.7.2

Bode plot 1: Constants and linear factors

7.2

Lathi 6.7.2

Analysis of damped 2nd order system

Video

We start by identifying what influences the position of poles in the transfer function.

We solve for the poles by finding the roots of the denominator of the transfer function.

Based on the **discriminant**, we have three outcomes:

1. Distinct and real roots.
2. Repeated roots
3. Complex conjugated root.

$$H(s) = \frac{b_0}{s^2 + a_1s + a_0} = \frac{b_0}{(s + p_1)(s + p_2)}$$

$$p_1 = \frac{a_1}{2} + \sqrt{\left(\frac{a_1}{2}\right)^2 - a_0}$$

$$p_2 = \frac{a_1}{2} - \sqrt{\left(\frac{a_1}{2}\right)^2 - a_0}$$

Poles:

$$s_1 = -p_1 = -\frac{a_1}{2} - \sqrt{\left(\frac{a_1}{2}\right)^2 - a_0}$$

$$s_2 = -p_2 = -\frac{a_1}{2} + \sqrt{\left(\frac{a_1}{2}\right)^2 - a_0}$$

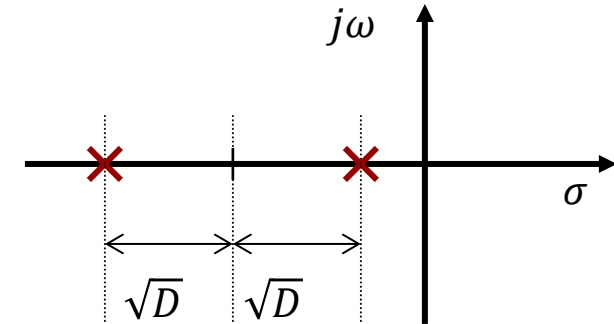
$$\text{Discriminant: } D = \left(\frac{a_1}{2}\right)^2 - a_0 \begin{cases} > 0 & \text{Overdamped} \\ = 0 & \text{Critically damped} \\ < 0 & \text{Underdamped} \end{cases}$$

Overdamped: $D > 0$

$$H(s) = \frac{C}{(s + p_1)(s + p_2)}$$

$$= \frac{A_1}{s + p_1} + \frac{A_2}{s + p_2}$$

$$h(t) = [A_1 e^{-p_1 t} + A_2 e^{-p_2 t}] u(t)$$

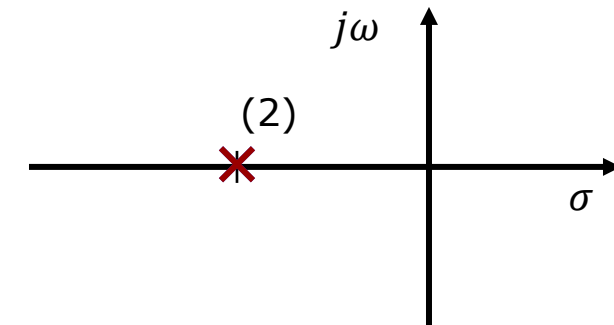


Critically damped: $D = 0$

$$H(s) = \frac{A}{(s + p)^2}$$

$$p = \frac{a_1}{2}$$

$$h(t) = A[t e^{-p t}] u(t)$$



Underdamped: $D < 0$

$$H(s) = \frac{b_0}{\underbrace{s^2 + a_1 s + a_0}_{\text{Mathematical form}}} = \frac{b_0}{a_0} \cdot \frac{\omega_n^2}{\underbrace{s^2 + 2\zeta\omega_n s + \omega_n^2}_{\text{Physical form}}}$$

Natural (undamped) frequency: $\omega_n = \sqrt{a_0}$

Damping ratio: $\zeta = \frac{a_1}{2\sqrt{a_0}}$

For underdamped systems, it is very useful to express the roots in terms of physically meaningful systems parameters, namely:

$$D = \left(\frac{a_1}{2}\right)^2 - a_0 = (\zeta\omega_n)^2 - \omega_n^2 = (\zeta^2 - 1)\omega_n^2$$

ζ : damping ratio

ω_n : the natural (undamped resonance frequency.

$$D \begin{cases} > 0 & \text{if } \zeta > 1: & \text{overdamped} \\ = 0 & \text{if } \zeta = 1: & \text{critically damped} \\ < 0 & \text{if } 0 < \zeta < 1: & \text{underdamped} \end{cases}$$

Damped Systems – pole positions expressed in terms of new systems parameters

Poles:

$$s_{1,2} = -p_{1,2} = -\frac{a_1}{2} \pm \sqrt{\left(\frac{a_1}{2}\right)^2 - a_0} = -\zeta\omega_n \pm \sqrt{(\zeta^2 - 1)\omega_n^2}$$

$$= -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2} = -\zeta\omega_n \pm j\omega_d$$

Damped frequency: ω_d

Observation:

The resonance frequency is influenced by the damping factor. The largest resonance frequency is for the completely undamped system ($\zeta = 0$). As $\zeta \rightarrow 1$ the damped frequency $\omega_d \rightarrow 0$.

Hence for critically damped and overdamped systems, there are no oscillations.

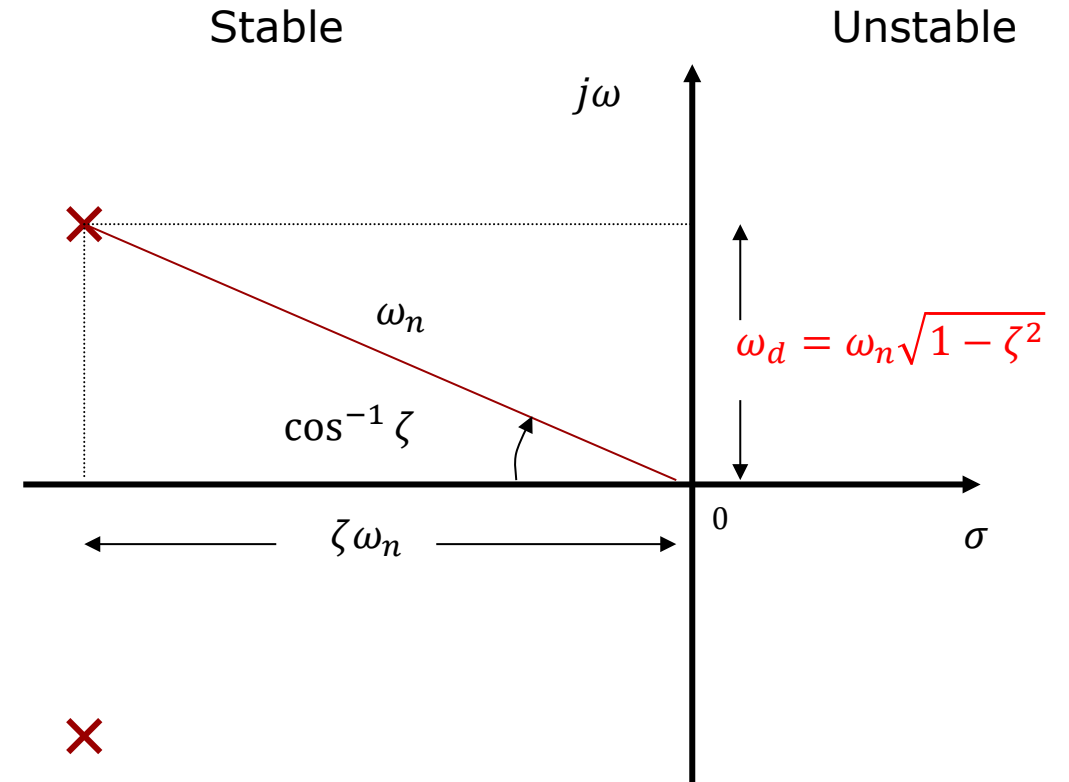
$$\zeta\omega_n = \omega_n \cos(\theta)$$

$$\zeta = \cos(\theta)$$

$$\zeta = 0 \Leftrightarrow \theta = 90^\circ$$

$$\zeta = \frac{1}{\sqrt{2}} \Leftrightarrow \theta = 45^\circ$$

$$\zeta = 1 \Leftrightarrow \theta = 0^\circ$$



Frequency Response – influence of damping factor

Expressing the transfer function $H(s)$ in terms of ζ and ω_n , we can now study the influence of these parameters on the **frequency response**.

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad H(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}$$

For very underdamped systems the amplitude spectrum features a local peak close to the 3dB cut-off frequency.

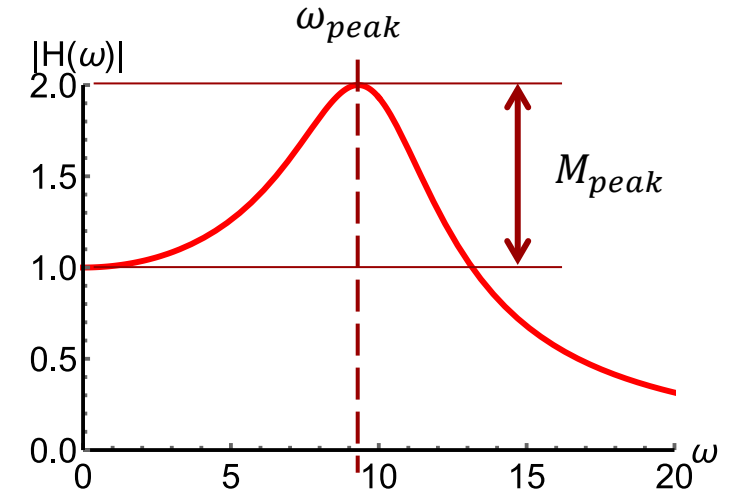
The frequency and the magnitude of the peak can be found by solving:

$$\frac{d|H(j\omega)|}{d\omega} = 0$$

Observation:

$\omega_{peak} \rightarrow \omega_n$ when $\zeta \rightarrow 0$

$\omega_{peak} \rightarrow 0$ when $\zeta \rightarrow 1/\sqrt{2}$



$$|H(j\omega)| = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}}$$

If $0 < \zeta < 1/\sqrt{2}$: $\omega_{peak} = \omega_n \sqrt{1 - 2\zeta^2}$

$$M_{peak} = \frac{1}{2\zeta\sqrt{1 - \zeta^2}}$$

Not in Lathi

$$\zeta \rightarrow \frac{1}{\sqrt{2}} \Rightarrow \omega_{peak} \rightarrow 0, M_{peak} \rightarrow 1$$

Frequency Response – influence of damping factor

No peak, but rolls off too early.

No peak, does not roll off too early.

Resonance peak

$$\zeta = 0.966 \quad \omega_n = 10$$

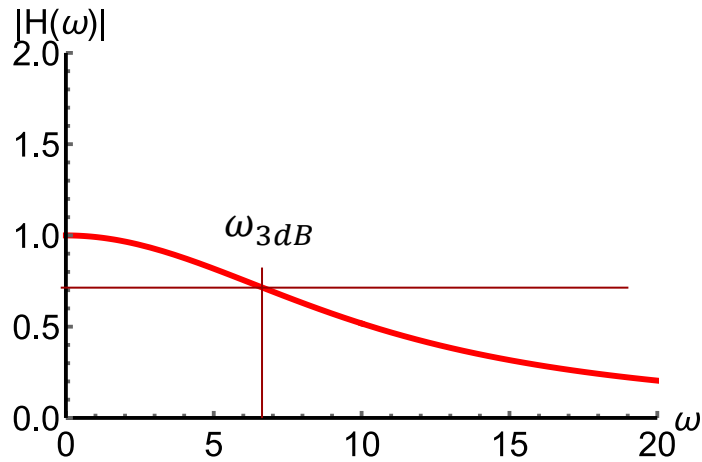
$$\theta = 15^\circ$$

$$\zeta = \frac{1}{\sqrt{2}} \approx 0.707 \quad \omega_n = 10$$

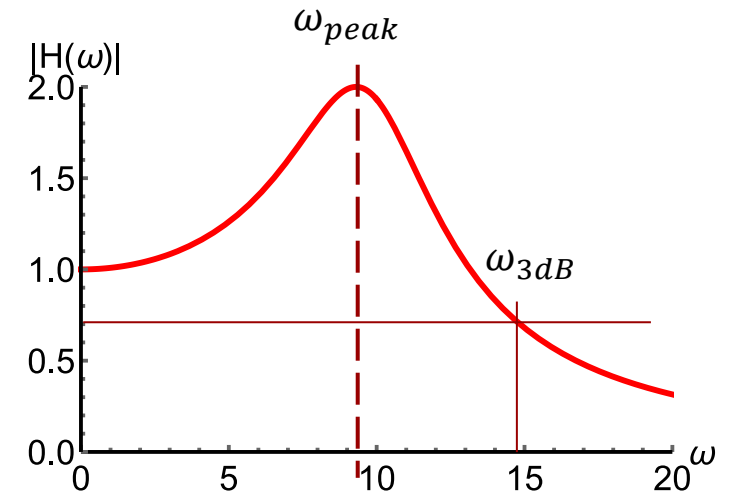
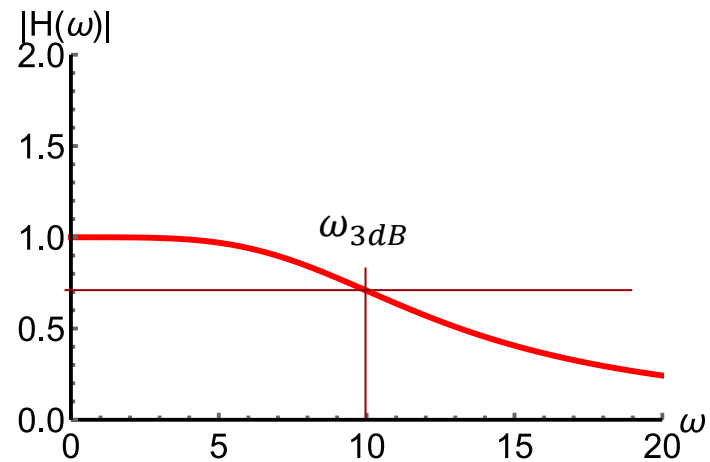
$$\theta = 45^\circ$$

$$\zeta = 0.259 \quad \omega_n = 10$$

$$\theta = 75^\circ$$



Close to critically damped



Close to undamped

Cut-off frequency dependence on damping factor

We would also like to know how the cut-off frequency ω_{3dB} varies as a function of damping ratio ζ .

We can get the modulus of the frequency from the transfer function:

We solve for that frequency ω_{3dB} where the pass-band gain has been dampened by a factor $\sqrt{2}$:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad H(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}$$

$$|H(j\omega_{3dB})| = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega_{3dB}^2)^2 + (2\zeta\omega_n\omega_{3dB})^2}} = \frac{1}{\sqrt{2}}$$

$$\omega_{3dB} = \sqrt{-2\zeta^2\omega_n^2 + \omega_n^2 + \sqrt{2} \sqrt{\omega_n^4(2\zeta^4 - 2\zeta^2 + 1)}}$$

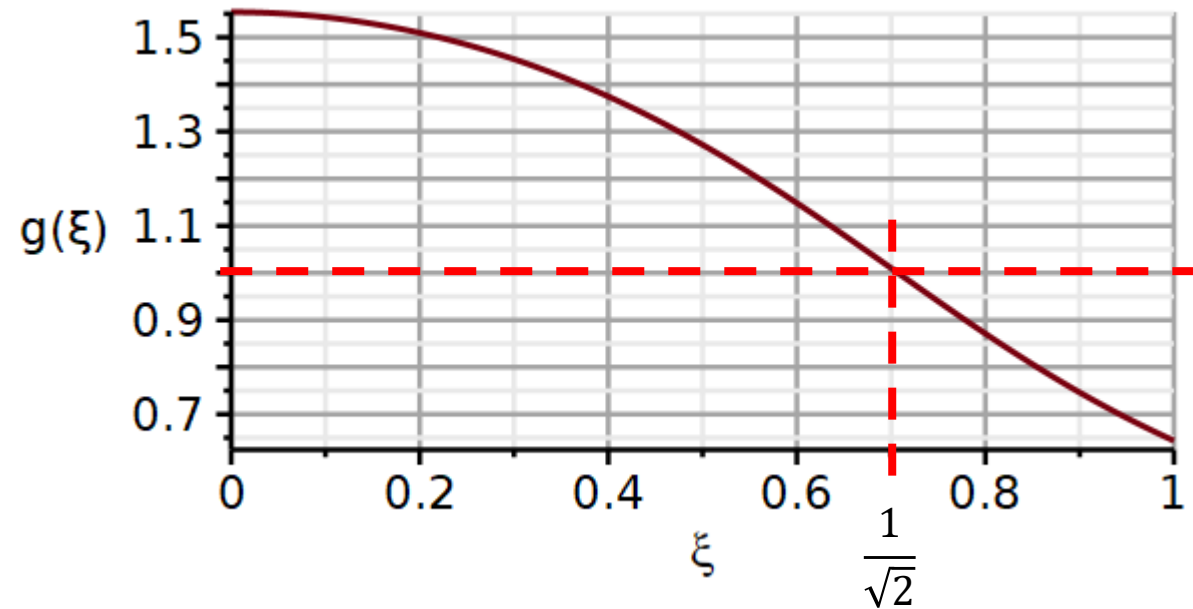
$$\omega_{3dB} = \omega_n \sqrt{-2\zeta^2 + 1 + \sqrt{2} \sqrt{(2\zeta^4 - 2\zeta^2 + 1)}}$$

$$\frac{\omega_{3dB}}{\omega_n} = \sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}$$

Cut-off frequency dependence on damping factor

$$\frac{\omega_{3dB}}{\omega_n} = \sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}$$

$$\frac{\omega_{3dB}}{\omega_n}$$



Observation:

The analysis tells us that the 3dB cut-off frequency for a 2nd order lowpass filter will increase if the damping factor is reduced while the undamped resonance frequency ω_n is kept unchanged.

$$\text{evalf}(g(1), 3)$$

0.640

$$\text{evalf}\left(g\left(\frac{1}{\sqrt{2}}\right), 3\right)$$

1.

$$\text{evalf}(g(0), 3)$$

1.55

Cut-off frequency dependence on damping factor

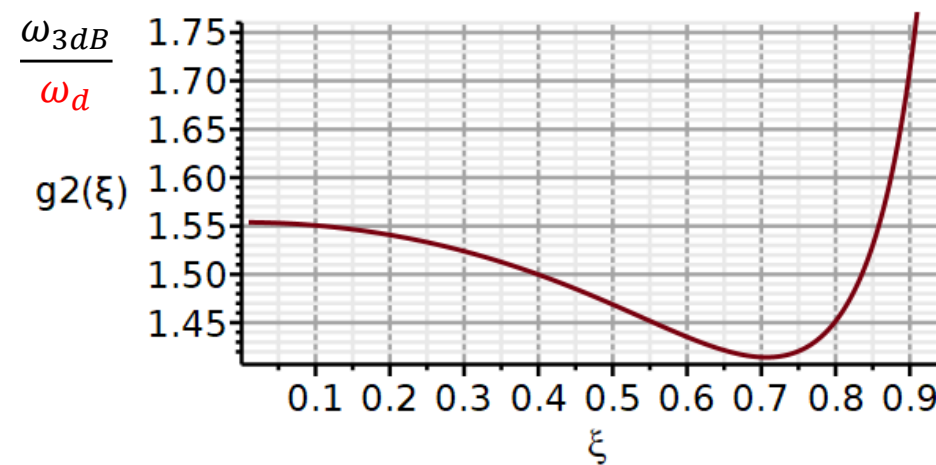
We can also find out how ω_{3dB} changes relative to ω_d $\omega_d = \omega_n \sqrt{1 - \zeta^2}$

$$\frac{\omega_{3dB}}{\omega_n} = \sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}$$

$$\frac{\omega_{3dB}}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}}{\sqrt{1 - \zeta^2}}$$

$$\frac{\omega_{3dB}}{\omega_d} = \frac{\sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}}{\sqrt{1 - \zeta^2}}$$

$$\frac{\omega_{3dB}}{\omega_d} = \sqrt{1 - \frac{\zeta^2}{(1 - \zeta^2)} + \frac{\sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}{(1 - \zeta^2)}}$$



$$\text{evalf}\left(g2\left(\frac{1}{\sqrt{2}}\right), 3\right)$$

1.41

$$\text{evalf}(g2(0), 3)$$

1.55

Cut-off frequency dependence on damping factor

The position of $\omega_{peak}, \omega_d, \omega_n, \omega_{3dB}$

$$\omega_n := 10 : \zeta := 0.1 ;$$

$$\omega_{3dB} := (\zeta, \omega_n) \rightarrow \omega_n \sqrt{1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}} :$$

$$\omega_d := (\zeta, \omega_n) \rightarrow \omega_n \cdot \sqrt{1 - \zeta^2} :$$

$$H0 := \omega \rightarrow \frac{\omega_n^2}{(j \cdot \omega)^2 + 2 \cdot \zeta \cdot \omega_n \cdot (j \cdot \omega) + \omega_n^2} :$$

$$Peak := [\omega_p, 20 \cdot \log_{10}(H_p(\zeta))]$$

$$Peak := [9.8995, 14.023]$$

$$H\omega_d := [\omega_d(\zeta, \omega_n), dB(H0, \omega_d(\zeta, \omega_n))]$$

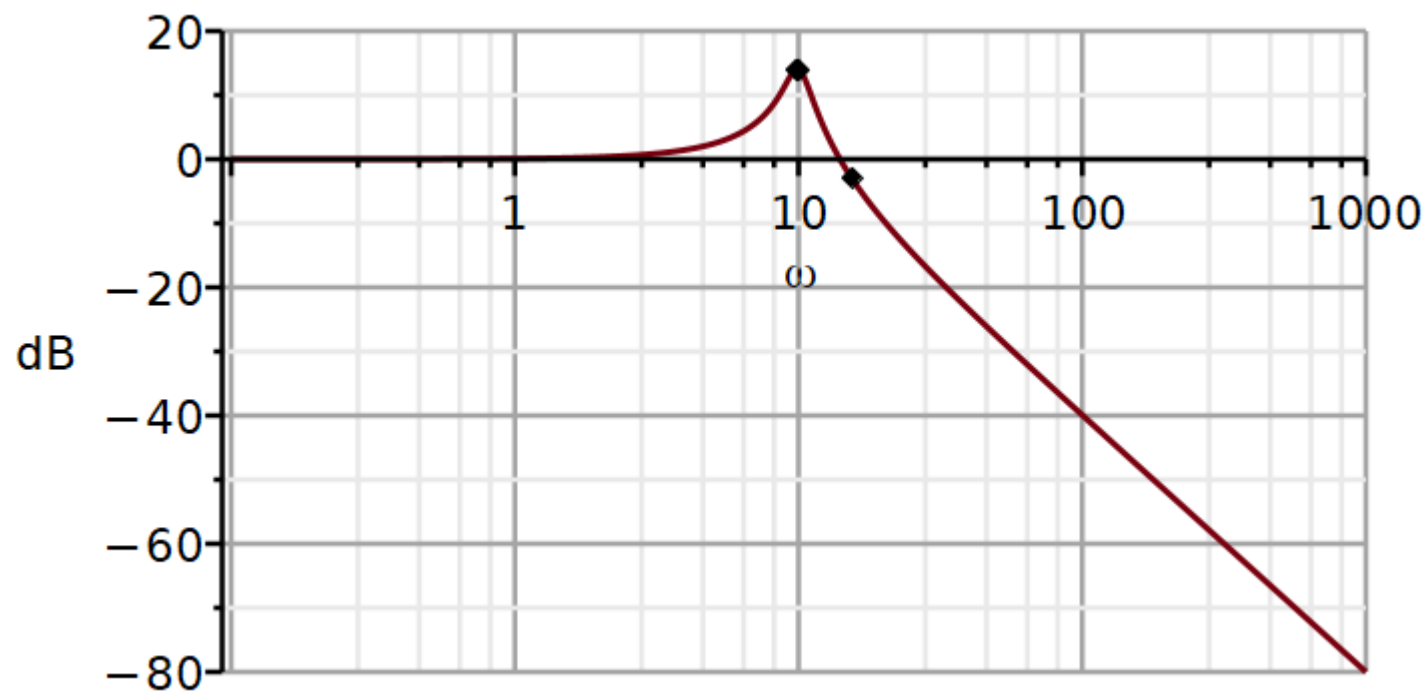
$$H\omega_d := [9.9499, 14.012]$$

$$Hn := [\omega_n, dB(H0, \omega_n)] ;$$

$$Hn := [10, 13.979]$$

$$H3dB := [\omega_{3dB}(\zeta, \omega_n), dB(H0, \omega_{3dB}(\zeta, \omega_n))]$$

$$H3dB := [15.428, -3.0108]$$



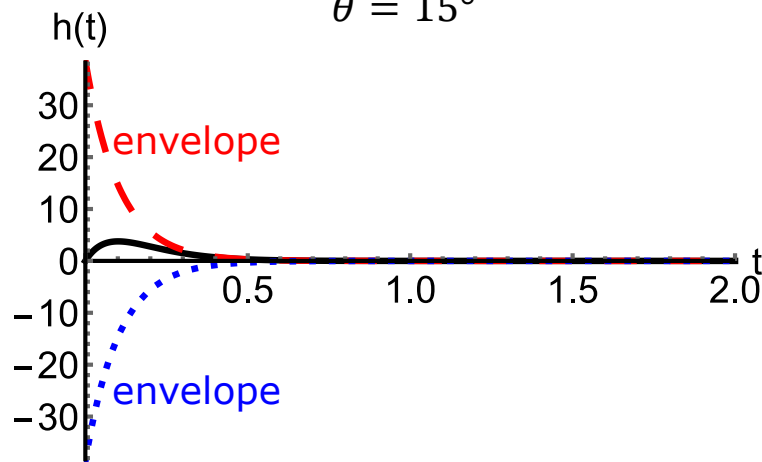
Time domain: Impulse response – influence of damping factor

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n}{\sqrt{1 - \zeta^2}} \cdot \frac{\omega_n \sqrt{1 - \zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

Table 6.1-9b: $h(t) = \underbrace{\frac{\omega_n}{\sqrt{1 - \zeta^2}}}_{\text{envelope}} \cdot e^{-\zeta\omega_n t} \sin \left[\underbrace{\left(\omega_n \sqrt{1 - \zeta^2} \right) t}_{\omega_d} \right] u(t), \tau = \frac{1}{\zeta\omega_n}$

$$\zeta = 0.966$$

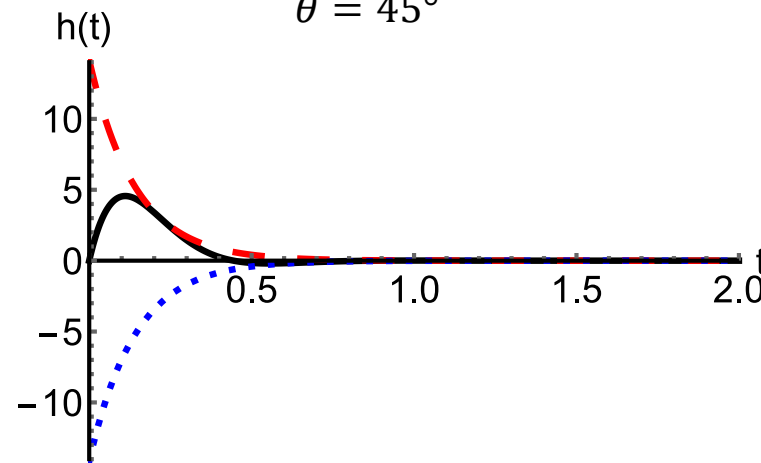
$$\theta = 15^\circ$$



Close to critically damped

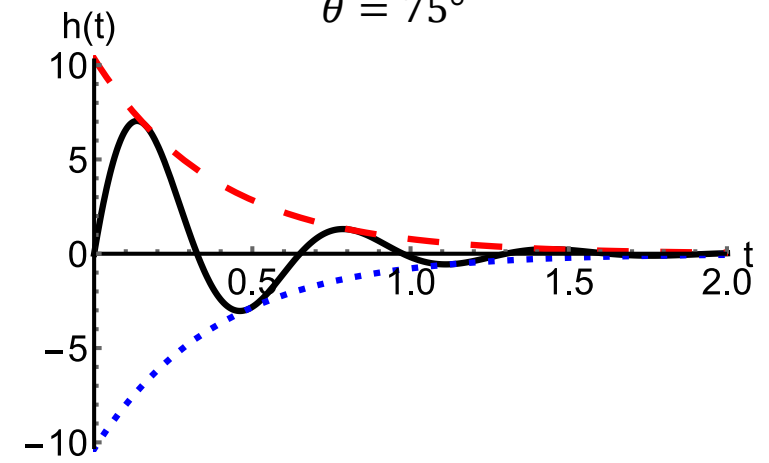
$$\zeta = 0.707$$

$$\theta = 45^\circ$$



$$\zeta = 0.259$$

$$\theta = 75^\circ$$



Close to undamped

How many periods in 4τ ?

$$h(t) = \underbrace{\frac{\omega_n}{\sqrt{1-\zeta^2}}}_{\text{envelope}} \cdot e^{-\zeta\omega_n t} \sin\left[\left(\omega_n\sqrt{1-\zeta^2}\right)t\right] u(t), \quad \tau = \frac{1}{\zeta\omega_n}$$

The sine function contains a frequency with a certain time-period T_d .

$$T_d = \frac{1}{f_d} = \frac{2\pi}{\omega_d} = \frac{2\pi}{\omega_n\sqrt{1-\zeta^2}}$$

The exponential function attenuates to with 2% in $t = 4\tau$

$$4\tau = \frac{4}{\zeta\omega_n}$$

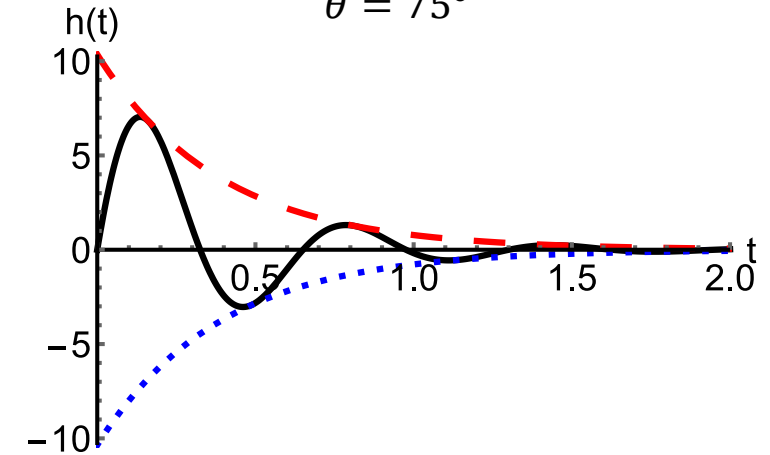
How many oscillations will occur within 4τ ?

We observe that it only depends on ζ .

For the example given in the graph, we get:

$$\zeta = 0.259$$

$$\theta = 75^\circ$$



$$\frac{4\tau}{T_d} = \frac{4}{\zeta\omega_n} \frac{\omega_n\sqrt{1-\zeta^2}}{2\pi} = \frac{4}{2\pi} \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$\left. \frac{4\tau}{T_d} \right|_{\zeta=0.259} = \frac{4}{2\pi} \times 3.73 = 2.37$$

$$\left. \frac{4\tau}{T_d} \right|_{\zeta=0.7070} = \frac{4}{2\pi} \times 1 = 0.637$$

Time domain: Step response – influence of damping factor

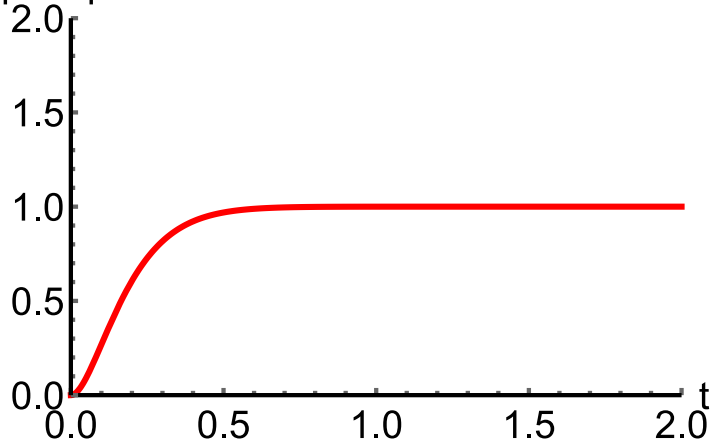
$$Y_{step}(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Table 6.1-10c: $y_{step}(t) = \left(1 - \frac{1}{\sqrt{1 - \zeta^2}} \cdot e^{-\zeta\omega_n t} \sin \left[\left(\omega_n \sqrt{1 - \zeta^2} \right) t + \cos^{-1}(\zeta) \right] \right) u(t)$

$$\zeta = 0.966$$

$$\theta = 15^\circ$$

step response

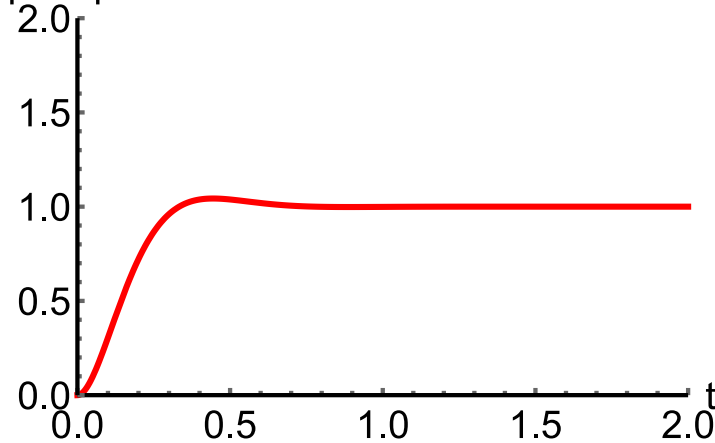


Close to critically damped

$$\zeta = 0.707$$

$$\theta = 45^\circ$$

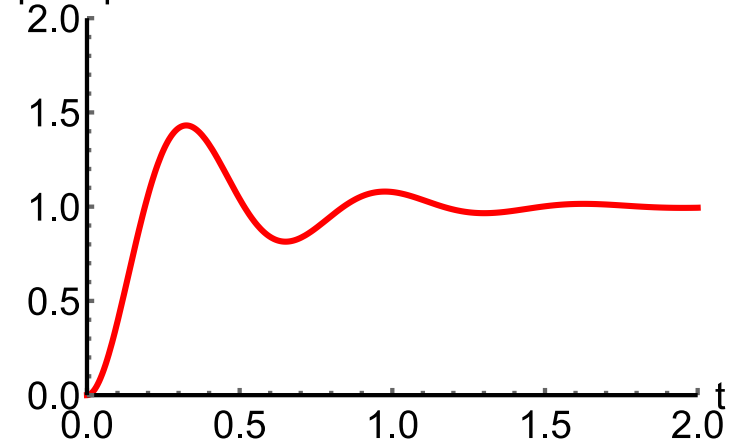
step response



$$\zeta = 0.259$$

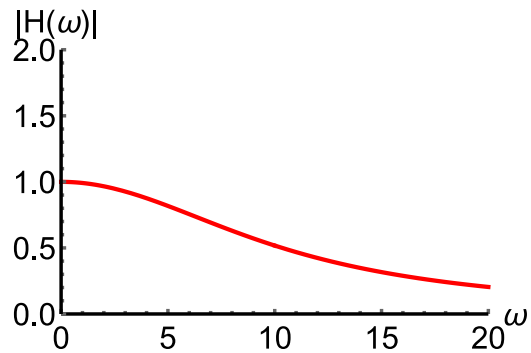
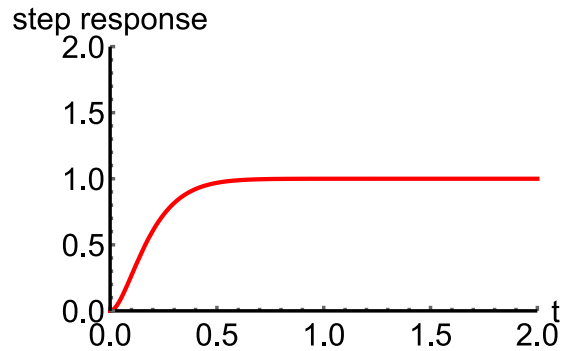
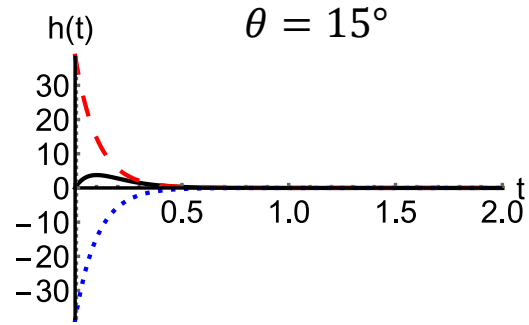
$$\theta = 75^\circ$$

step response

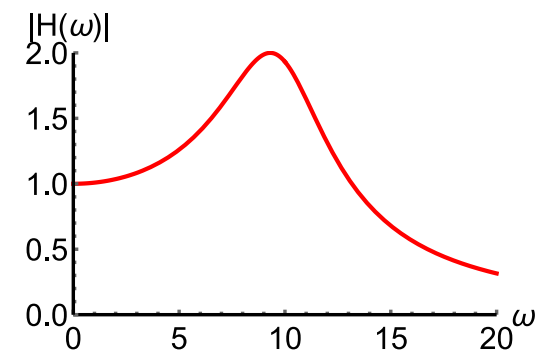
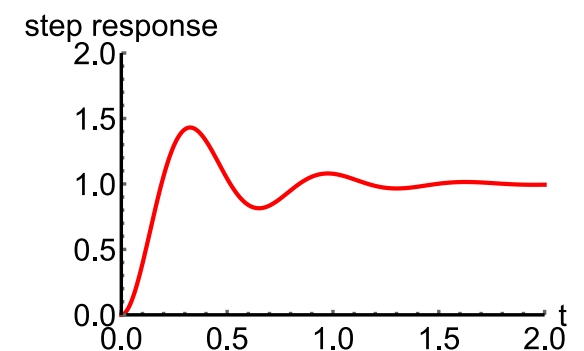
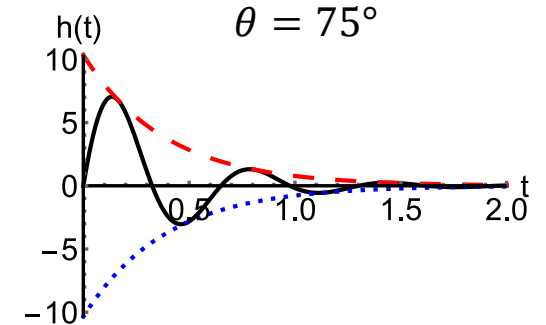
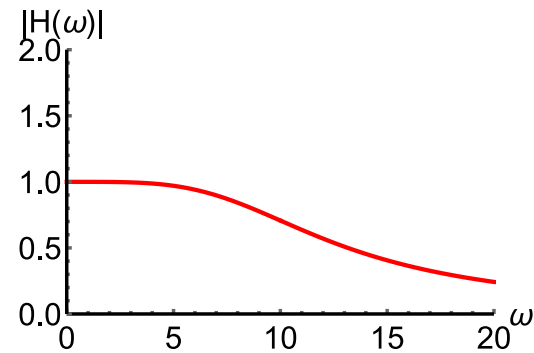
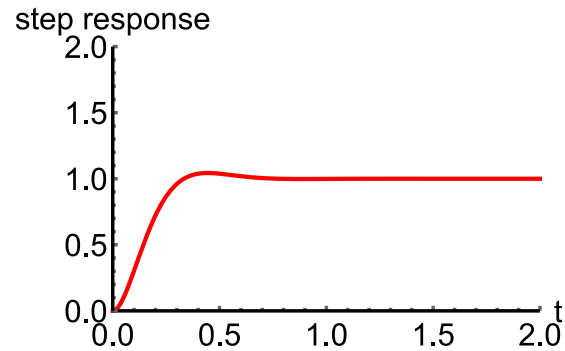
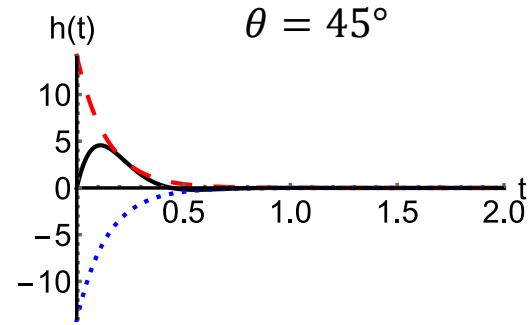


Close to undamped

Response for various degrees of damping



Close to critically damped



Close to undamped

Step response features

Video

We have seen on the previous slides, that different system parameters yield different system responses.

Can we flip this around and determine system parameters from systems response curves?

If we could, then we would be able to conduct an experiment with an unknown system, apply a step input, determine the system's response and from that write the differential equation for the system.

To do so, we need to define quantitative features of the system response. With these definitions in place, we may be able to determine what values the system parameters need to have to meet system performance criteria expressed in terms of quantitative time domain response features.

One set of response features are the **response times** measurable on the step response.

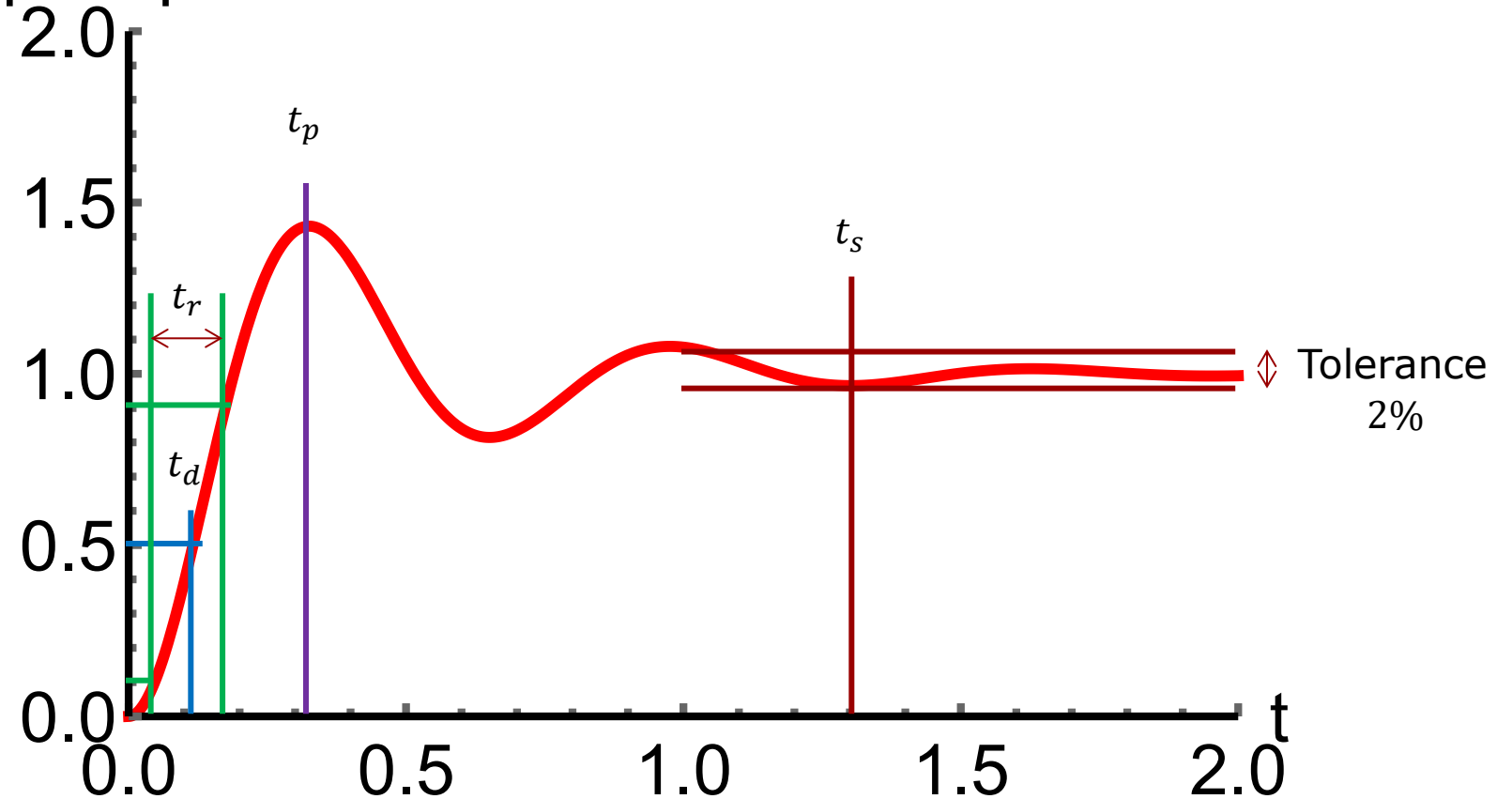
step response

Rise time (10 to 90%): t_r

Delay time(time to 50%: t_d

Time to peak: t_p

2% settling time: t_s



Response times derived from the analytical expression for the step response.

$$y_{step}(t) = \left(1 - \frac{1}{\sqrt{1 - \zeta^2}} \cdot e^{-\zeta \omega_n t} \sin \left[\left(\omega_n \sqrt{1 - \zeta^2} \right) t + \cos^{-1}(\zeta) \right] \right) u(t)$$

2% settling time: $t_s = 4\tau = \frac{4}{\zeta \omega_n}$

Time to peak: $t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_d} = \frac{\pi}{2\pi f_d} = \frac{T_d}{2}$

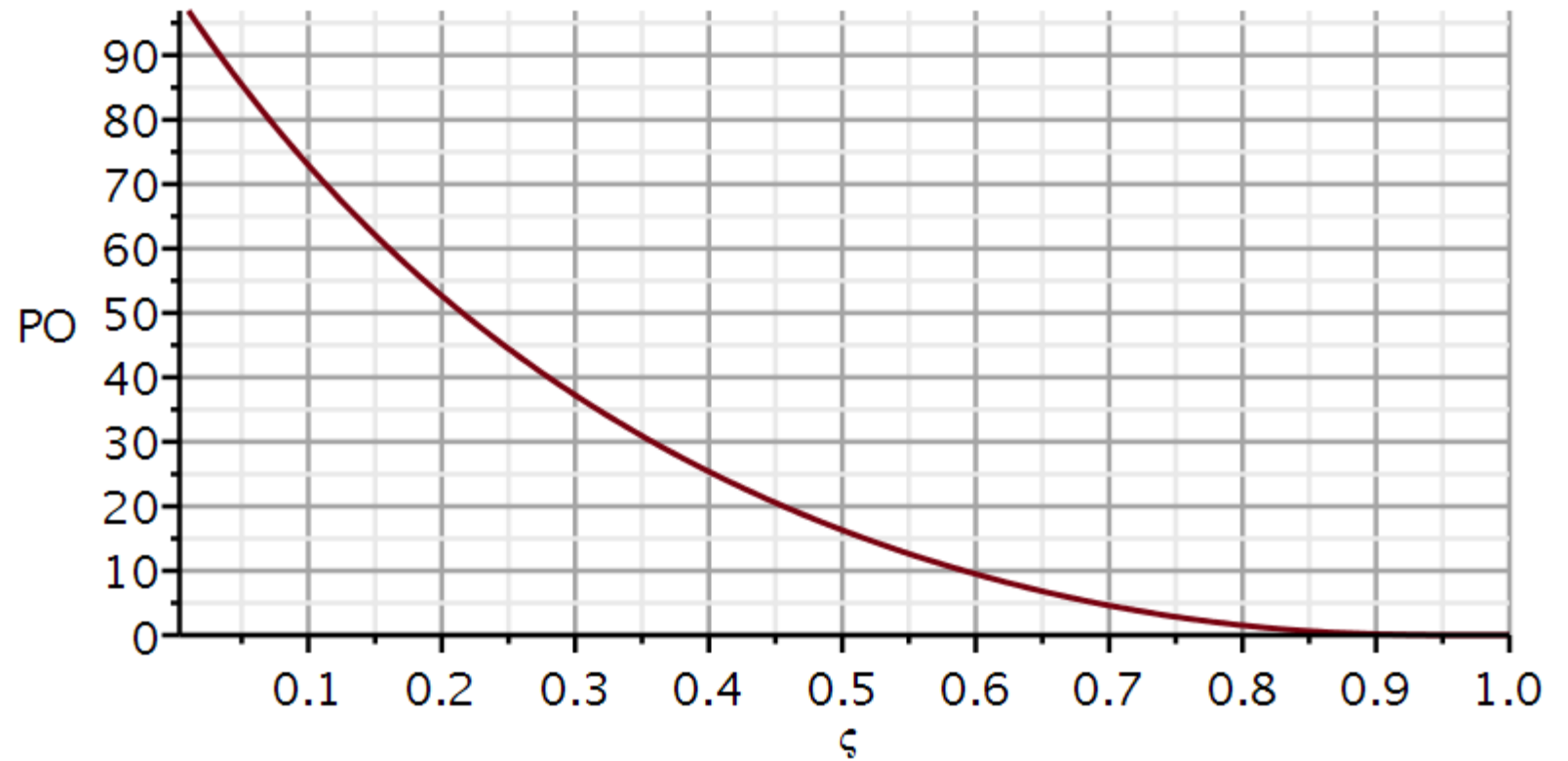
% overshoot: $PO = \frac{y(t_p) - y_\infty}{y_\infty} \cdot 100\% = 100e^{-\zeta \pi / \sqrt{1 - \zeta^2}}$

Rise time: $t_r \approx \frac{1 - 0.4167\zeta + 2.917\zeta^2}{\omega_n}$

Delay time: $t_d \approx \frac{1.1 + 0.125\zeta + 0.469\zeta^2}{\omega_n}$

Estimating damping ratio from percent overshoot

The damping factor ζ can be determined independently from the **percent overshoot**.



Approach: $PO \rightarrow \zeta; t_p \rightarrow \omega_n$
$$PO = \frac{y(t_p) - y_\infty}{y_\infty} \cdot 100\% = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

Step 1: Read PO on response curve and convert to percentage.

Step 2: Calculate damping ratio: $a \stackrel{\text{def}}{=} \ln\left(\frac{PO}{100}\right) \rightarrow \zeta = \frac{|a|}{\sqrt{\pi^2 + a^2}}$

Step 3: Read time to peak on response curve: t_p

Step 4: Calculate:
$$\omega_n = \frac{\pi}{t_p \sqrt{1 - \zeta^2}}$$

Remember: This algorithm is only applicable to a second order **underdamped** system.

Example: System Identification using quantitative features of the step response

Write the differential equation for this system.

$PO \approx 72\%$, NOT 172%

→ $\zeta \approx 0.104$

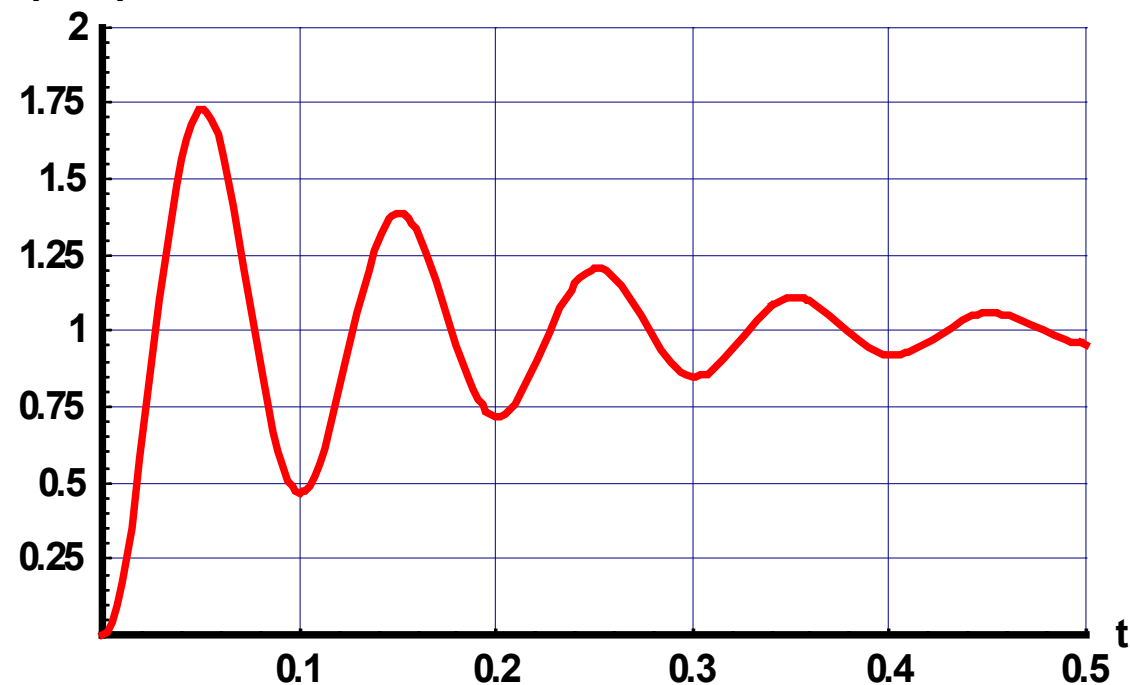
$t_p \approx 0.05$

$\omega_n \approx 63.2$

$a_0 = \omega_n^2 \approx 3990$

$a_1 = 2\zeta\omega_n \approx 13.1$

step response



$$\ddot{y}(t) + 13.1\dot{y}(t) + 3990y(t) = 3990x(t)$$

What about the right-hand side?

We have seen that, in case of the underdamped system, the system response and its frequency characteristics are all determined by two parameters: ζ and ω_n , which again are determined from the positions of the complex conjugated poles.

It is therefore a very useful skill to be able to anticipate changes in systems response and frequency characteristics caused by simple displacements of the poles.

The many plots on the next slide illustrate what happens to the impulse response, the step response, and the amplitude characteristics, when we move the poles along certain simple trajectories.

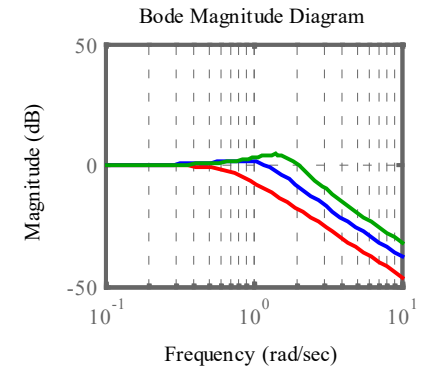
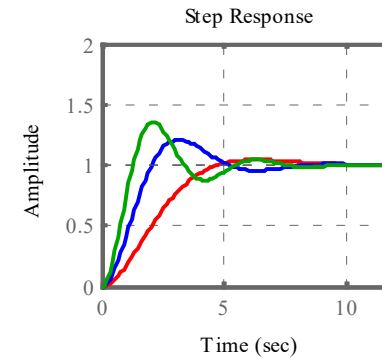
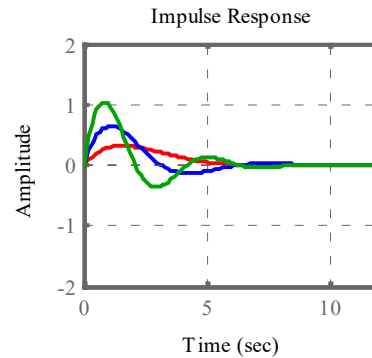
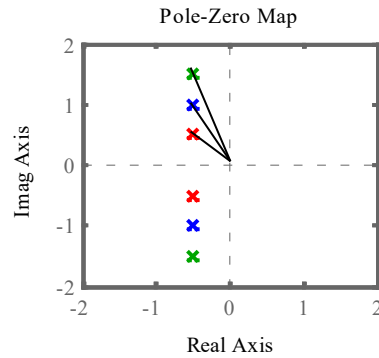
Try and see, if you can make heads and tails of the interconnections between plots.

Hint. Try and figure out if moving the poles along a certain trajectory makes the system more or less dampened.

Pole Influence on Dynamic System Characteristics

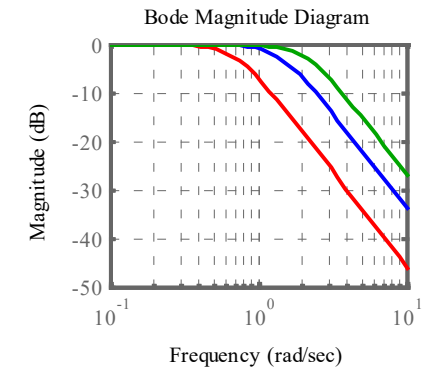
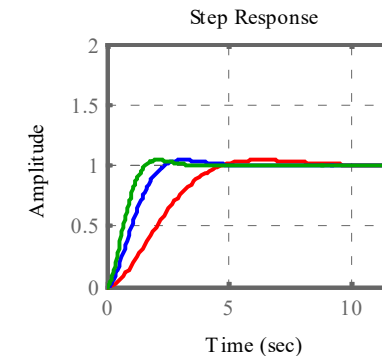
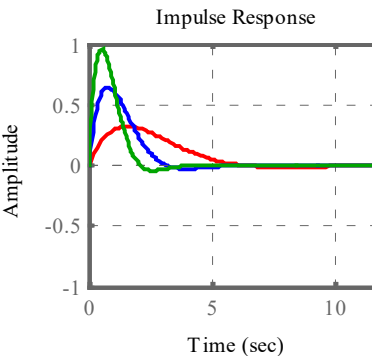
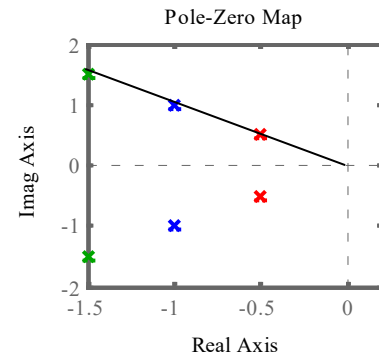
What is common:

Constant $\zeta\omega_n$

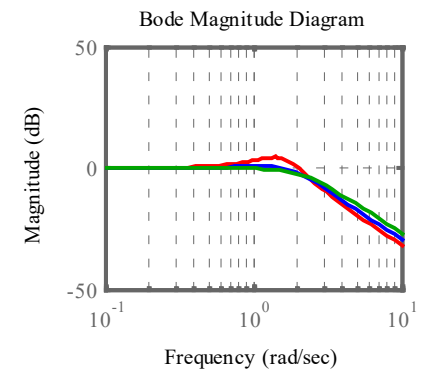
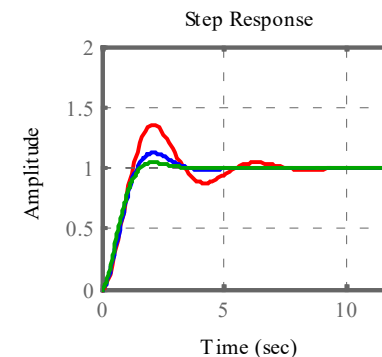
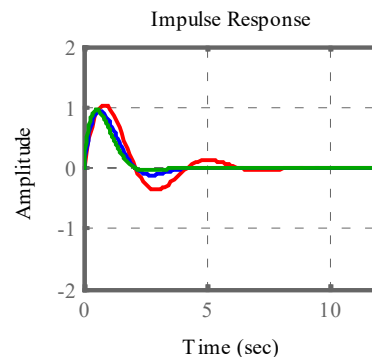
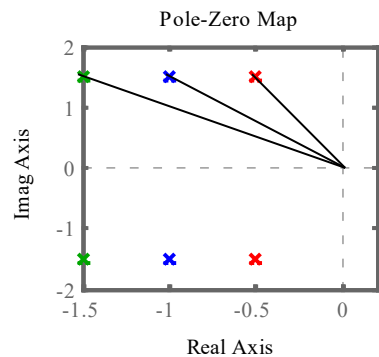


Constant θ

Constant ζ



Constant ω_d



Maple feature in DynamicSystems

Maple's DynamicSystems can calculate features on the step response.

Here we do it ourself first for comparison.

$$\begin{aligned}
 s_{1,2} &= -p_{1,2} = -\frac{a_1}{2} \pm \sqrt{\left(\frac{a_1}{2}\right)^2 - a_0} = -\zeta\omega_n \pm \sqrt{(\zeta^2 - 1)\omega_n^2} \\
 &= -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2} = -\zeta\omega_n \pm j\omega_d
 \end{aligned}$$

Dynamic systems 1

```

restart
with(plots) :
with(DynamicSystems) :
vin := Step(1, 0, 0) :
p1 := -1 + j·1 ;; p2 := -1 - j·1 ;;
expand((s - p1)·(s - p2))

```

$$s^2 + 2s + 2$$

```
a1 := -2·Re(p1)
```

$$a1 := 2$$

```
a0 := (Re(p1))^2 + (Im(p1))^2
```

$$a0 := 2$$

```

ζ := evalf( (a1 / (2·sqrt(a0))) )

```

$$\zeta := 0.7071067810$$

```
ωn := evalf(sqrt(a0))
```

$$\omega_n := 1.414213562$$

```
ts := evalf( 4 / (ζ·ωn) )
```

$$t_s := 4.000000004$$

$$t_p := \text{evalf}\left(\frac{\pi}{\omega_n \cdot \sqrt{1 - \zeta^2}}\right)$$

$$t_p := 3.141592654$$

$$PO := 100 \cdot e^{-\zeta \cdot \frac{\pi}{\sqrt{1 - \zeta^2}}}$$

$$PO := 4.321391829$$

$$t_r := \text{evalf}\left(\frac{1 - 0.5167 \cdot \zeta + 2.917 \cdot \zeta^2}{\omega_n}\right)$$

$$t_r := 1.480072021$$

$$t_d := \text{evalf}\left(\frac{1.1 + 0.125 \cdot \zeta + 0.469 \cdot \zeta^2}{\omega_n}\right)$$

$$t_d := 1.006134000$$

Now define the system:

$$Ts := 0.05 :$$

$$dur := 10 :$$

$$Ns := \text{round}\left(\frac{dur}{Ts}\right) :$$

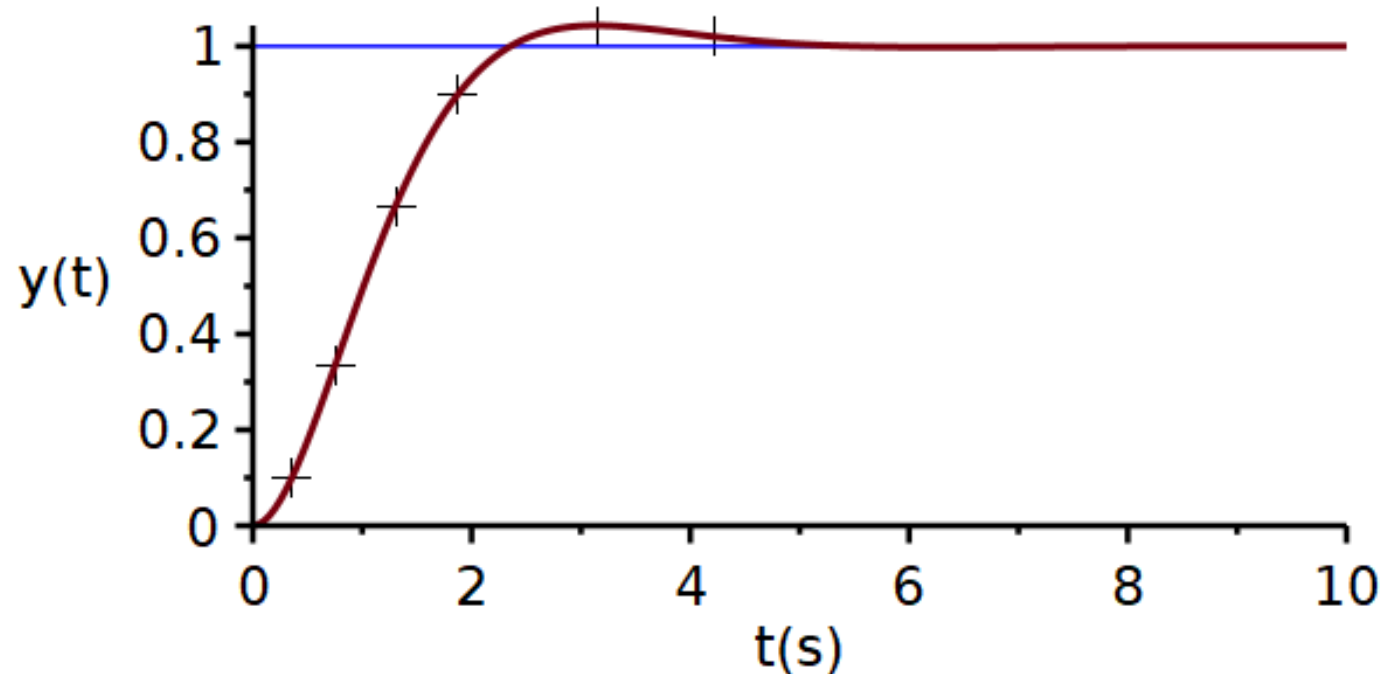
$$sys1 := \text{TransferFunction}\left(\frac{(\omega_n)^2}{s^2 + 2 \cdot \zeta \cdot \omega_n \cdot s + (\omega_n)^2}\right) :$$



Using the function StepProperties:

```
prop := StepProperties(sys1) :
prop
1., [ 0.357403497705475, 0.100000000000000], [ 0.751462782947410, 0.333333333333333],
    [ 1.30413573538299, 0.666666666666667], [ 1.87629572625499, 0.900000000000000],
    [ 3.14159265465485, 1.04321391828416], [ 4.21618403228512, 1.020000000000000]
Final value
prop[ 1]
1.
10% of final value
prop[ 2]
[ 0.357403497705475, 0.100000000000000]
33% of final value
prop[ 3]
[ 0.751462782947410, 0.333333333333333]
67% of final value
prop[ 4]
[ 1.30413573538299, 0.666666666666667]
90% of final value
prop[ 5]
[ 1.87629572625499, 0.900000000000000]   tr := 1.480072021
peak value
prop[ 6]
[ 3.14159265465485, 1.04321391828416]   tp := 3.141592654
Settling value
prop[ 7]
[ 4.21618403228512, 1.020000000000000]   ts := 4.000000004
```

```
plots[display](plot([prop[2..7]], style=point, symbol=cross, color=black, symbolsize=30),
  plot([ [0, prop[1]], [dur, prop[1]]], color=blue), ResponsePlot(sys1, Step(), duration=dur,
  numpoints=200, thickness=3.0), axis[2]=[thickness=2.5], axis[1]=[mode=linear, thickness
  =2.5], font=[Helvetica, roman, 18], labels=["t(s)", "y(t)"], labelfont=["HELVETICA", 18],
  size=[600, 300])
```



Here the step response is plotted together with the measured features marked as a "+".

Bode Plot 1:

Conversion of $H(s)$ to Bode plot form

Video

The transfer function can be written as **rational functions**, i.e., as a ratio of two polynomials.

$$H(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

The numerator and denominator can be factored:

$$H(s) = K \frac{(s + b_1)(s + b_2) \dots (s + b_m)}{(s + a_1)(s + a_2) \dots (s + a_n)}$$

There may be zeros and poles at the origin:

$$H(s) = K \frac{s^p (s + b_1)(s + b_2) \dots (s + b_{m-p})}{s^q (s + a_1)(s + a_2) \dots (s + a_{n-q})}$$

We can combine all gain factors in one:

$$H(s) = K \frac{b_1 b_2 \dots b_{m-p}}{a_1 a_2 \dots a_{n-q}} \frac{s^p \left(\frac{s}{b_1} + 1\right) \left(\frac{s}{b_2} + 1\right) \dots \left(\frac{s}{b_{m-p}} + 1\right)}{s^q \left(\frac{s}{a_1} + 1\right) \left(\frac{s}{a_2} + 1\right) \dots \left(\frac{s}{a_{n-q}} + 1\right)}$$

Complex conjugated factors are kept as quadratic factors:

$$H(s) = K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} \frac{s^p \left(\frac{s}{b_1} + 1\right) \left(\frac{s}{b_2} + 1\right)^{m_2} \left(\frac{s^2}{b_4} + \frac{b_3}{b_4} s + 1\right)^{m_3}}{s^q \left(\frac{s}{a_1} + 1\right) \left(\frac{s}{a_2} + 1\right)^{n_2} \left(\frac{s^2}{a_4} + \frac{a_3}{a_4} s + 1\right)^{n_3}}$$

The s-notation form is valid all over the complex number plane:

$$H(s) = K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} \frac{s^p \left(\frac{s}{b_1} + 1\right) \left(\frac{s}{b_2} + 1\right)^{m_2} \left(\frac{s^2}{b_4} + \frac{b_3}{b_4} s + 1\right)^{m_3}}{s^q \left(\frac{s}{a_1} + 1\right) \left(\frac{s}{a_2} + 1\right)^{n_2} \left(\frac{s^2}{a_4} + \frac{a_3}{a_4} s + 1\right)^{n_3}}$$

We need the frequency characteristic and insert $j\omega \rightarrow s$

$$H(j\omega) = K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} \frac{(j\omega)^p \left(\frac{j\omega}{b_1} + 1\right) \left(\frac{j\omega}{b_2} + 1\right)^{m_2} \left(\frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1\right)^{m_3}}{(j\omega)^q \left(\frac{j\omega}{a_1} + 1\right) \left(\frac{j\omega}{a_2} + 1\right)^{n_2} \left(\frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1\right)^{n_3}}$$

We need to plot the amplitude and phase spectra in separate plots:

$$|H(j\omega)| = K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} \frac{|j\omega|^p \left|\frac{j\omega}{b_1} + 1\right| \left|\frac{j\omega}{b_2} + 1\right|^{m_2} \left|\frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1\right|^{m_3}}{|j\omega|^q \left|\frac{j\omega}{a_1} + 1\right| \left|\frac{j\omega}{a_2} + 1\right|^{n_2} \left|\frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1\right|^{n_3}}$$

$$\begin{aligned} \angle H(\omega) = & p\angle(j\omega) + \angle\left(\frac{j\omega}{b_1} + 1\right) + m_2\angle\left(\frac{j\omega}{b_2} + 1\right) + m_3\angle\left(\frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1\right) \\ & - q\angle(j\omega) - \angle\left(\frac{j\omega}{a_1} + 1\right) - n_2\angle\left(\frac{j\omega}{a_2} + 1\right) - n_3\angle\left(\frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1\right) \end{aligned}$$

Amplitude characteristic:

$$|H(j\omega)| = K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} \frac{|j\omega|^p \left| \frac{j\omega}{b_1} + 1 \right| \left| \frac{j\omega}{b_2} + 1 \right|^{m_2} \left| \frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1 \right|^{m_3}}{|j\omega|^q \left| \frac{j\omega}{a_1} + 1 \right| \left| \frac{j\omega}{a_2} + 1 \right|^{n_2} \left| \frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1 \right|^{n_3}}$$

We convert amplitude to decibel and make use of:

$$20 \log_{10} \left(\frac{x \cdot y^m}{z^n} \right) = 20 \log_{10} x + m \cdot 20 \log_{10} y - n \cdot 20 \log_{10} z$$

Now each factor has become a term:

$$\begin{aligned} |H(j\omega)|_{dB} = & 20 \log_{10} K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} + p \times 20 \log_{10} |j\omega| + 20 \log_{10} \left| \frac{j\omega}{b_1} + 1 \right| \\ & + m_2 \times 20 \log_{10} \left| \frac{j\omega}{b_2} + 1 \right| + m_3 \times 20 \log_{10} \left| \frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1 \right| \\ & - q \times 20 \log_{10} |j\omega| - 20 \log_{10} \left| \frac{j\omega}{a_1} + 1 \right| - n_2 \times 20 \log_{10} \left| \frac{j\omega}{a_2} + 1 \right| \\ & - n_3 \times 20 \log_{10} \left| \frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1 \right| \end{aligned}$$

Standard form for Bode plot

The advantage of taking $\log_{10}$ is that each factor now becomes a term. We can now draw an amplitude and phase curve for each factor, and then we can add the curves together graphically.

As each term may originate from a specific sub-system, we can easily identify the behavior of each sub-system and then correct those where needed.

Furthermore, if we decide to add a correcting subsystem into our system, we just add its Bode plot to the Bode plot of the original system.

$$\begin{aligned}
 |H(j\omega)|_{dB} = & \underbrace{20 \log_{10} K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}}}_{\text{constant}} + \underbrace{p \times 20 \log_{10} |j\omega|}_{\text{zero at origin}} + \underbrace{20 \log_{10} \left| \frac{j\omega}{b_1} + 1 \right|}_{\text{first order zero}} \\
 & + \underbrace{m_2 \times 20 \log_{10} \left| \frac{j\omega}{b_2} + 1 \right|}_{\text{repeated first order zero}} + \underbrace{m_3 \times 20 \log_{10} \left| \frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1 \right|}_{\text{repeated second order zero}} \\
 & - \underbrace{q \times 20 \log_{10} |j\omega|}_{\text{pole at origin}} - \underbrace{20 \log_{10} \left| \frac{j\omega}{a_1} + 1 \right|}_{\text{first order pole}} - \underbrace{n_2 \times 20 \log_{10} \left| \frac{j\omega}{a_2} + 1 \right|}_{\text{repeated first order pole}} \\
 & - \underbrace{n_3 \times 20 \log_{10} \left| \frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1 \right|}_{\text{repeated second order pole}}
 \end{aligned}$$

$$\begin{aligned}
 \angle H(\omega) = & p \angle(j\omega) + \angle \left(\frac{j\omega}{b_1} + 1 \right) + m_2 \angle \left(\frac{j\omega}{b_2} + 1 \right) + m_3 \angle \left(\frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1 \right) \\
 & - q \angle(j\omega) - \angle \left(\frac{j\omega}{a_1} + 1 \right) - n_2 \angle \left(\frac{j\omega}{a_2} + 1 \right) - n_3 \angle \left(\frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1 \right)
 \end{aligned}$$

Example 1

Rewrite transfer
function into
Bode plot form

$$H(s) = 10 \frac{s(s+1)}{(s+2)(s^2+2s+100)^2} = 10 \cdot \frac{1}{2 \cdot 100^2} \cdot \frac{\frac{s}{1} \left(\frac{s}{1} + 1 \right)}{\left(\frac{s}{2} + 1 \right) \left(\left(\frac{s}{10} \right)^2 + \frac{2}{10} \frac{s}{10} + 1 \right)^2}$$

$$H(j\omega) = 5 \cdot 10^{-3} \cdot \frac{j\omega}{1} \cdot \left(\frac{j\omega}{1} + 1 \right) \cdot \left(\frac{j\omega}{2} + 1 \right)^{-1} \cdot \left(\left(\frac{j\omega}{10} \right)^2 + \frac{2}{10} \frac{j\omega}{10} + 1 \right)^{-2}$$

$$\begin{aligned} |H(j\omega)|_{dB} = & 20 \log_{10} 5 \cdot 10^{-3} + 20 \log_{10} \frac{j\omega}{1} + 20 \log_{10} \left(\frac{j\omega}{1} + 1 \right) \\ & - 1 \cdot 20 \log_{10} \left(\frac{j\omega}{2} + 1 \right) - 2 \cdot 20 \log_{10} \left(\left(\frac{j\omega}{10} \right)^2 + \frac{2}{10} \frac{j\omega}{10} + 1 \right) \end{aligned}$$

$$\angle H(\omega) = \angle(j\omega) + \angle \left(\frac{j\omega}{1} + 1 \right) - \angle \left(\frac{j\omega}{2} + 1 \right) - 2 \angle \left(\left(\frac{j\omega}{10} \right)^2 + \frac{2}{10} \frac{j\omega}{10} + 1 \right)$$

Example 2

Rewrite into
standard
form for Bode
Plot

$$H(s) = 50 \frac{(s+5)(s^2+s+25)}{s(s+4)^2(s^2+10s+100)} = 50 \cdot \frac{5 \cdot 25}{4^2 \cdot 100} \cdot \frac{\left(\frac{s}{5} + 1\right) \left(\left(\frac{s}{5}\right)^2 + \frac{1}{5} \frac{s}{5} + 1\right)}{\frac{s}{1} \left(\frac{s}{4} + 1\right)^2 \left(\left(\frac{s}{10}\right)^2 + \frac{10}{10} \frac{s}{10} + 1\right)}$$

$$H(j\omega) = 50 \cdot \frac{5 \cdot 25}{4^2 \cdot 100} \cdot \left(\frac{j\omega}{5} + 1\right) \left(\left(\frac{j\omega}{5}\right)^2 + \frac{1}{5} \frac{j\omega}{5} + 1\right) \left(\frac{j\omega}{1}\right)^{-1} \left(\frac{j\omega}{4} + 1\right)^{-2} \left(\left(\frac{j\omega}{10}\right)^2 + \frac{10}{10} \frac{j\omega}{10} + 1\right)^{-1}$$

$$|H(j\omega)|_{dB} = 20 \log_{10} \left(50 \cdot \frac{5 \cdot 25}{4^2 \cdot 100} \right) + 20 \log_{10} \left(\frac{j\omega}{5} + 1 \right) + 20 \log_{10} \left(\left(\frac{j\omega}{5}\right)^2 + \frac{1}{5} \frac{j\omega}{5} + 1 \right) \\ - 20 \log_{10} \left(\frac{j\omega}{1} \right) - 2 \cdot 20 \log_{10} \left(\frac{j\omega}{4} + 1 \right) - 20 \log_{10} \left(\left(\frac{j\omega}{10}\right)^2 + \frac{10}{10} \frac{j\omega}{10} + 1 \right)$$

$$\angle H(\omega) = \angle \left(\frac{j\omega}{5} + 1 \right) + \angle \left(\left(\frac{j\omega}{5}\right)^2 + \frac{1}{5} \frac{j\omega}{5} + 1 \right) - \angle \left(\frac{j\omega}{1} \right) - 2 \angle \left(\frac{j\omega}{4} + 1 \right) - \angle \left(\left(\frac{j\omega}{10}\right)^2 + \frac{10}{10} \frac{j\omega}{10} + 1 \right)$$

Drawing Bode factors

Constants and linear factors

Video

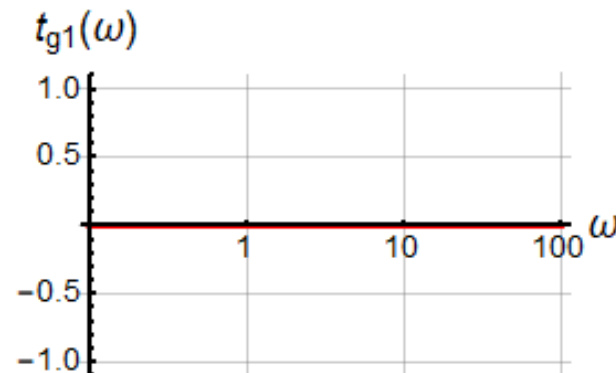
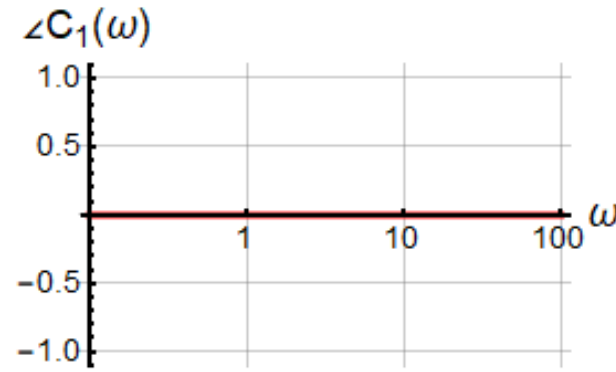
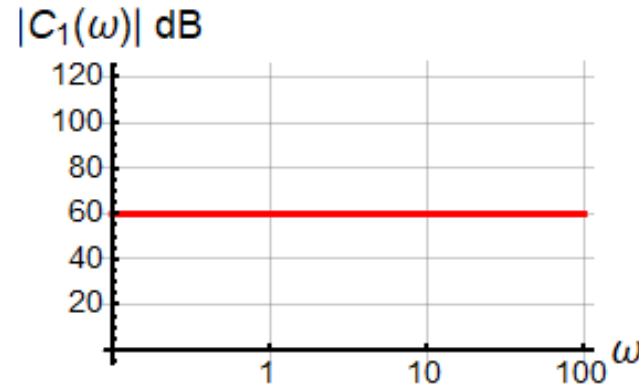
Constant Factor

$$|C(\omega)|_{dB} = 20 \cdot \log \left(K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} \right)$$

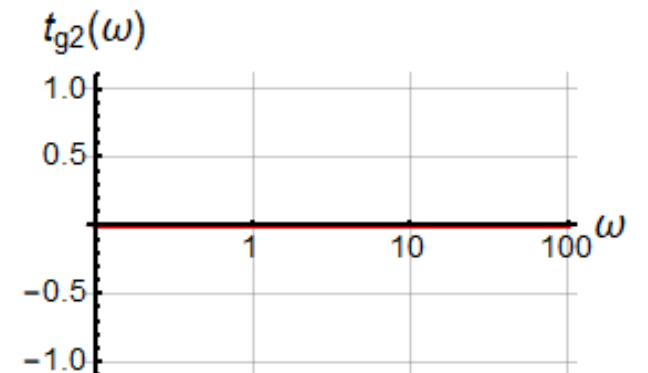
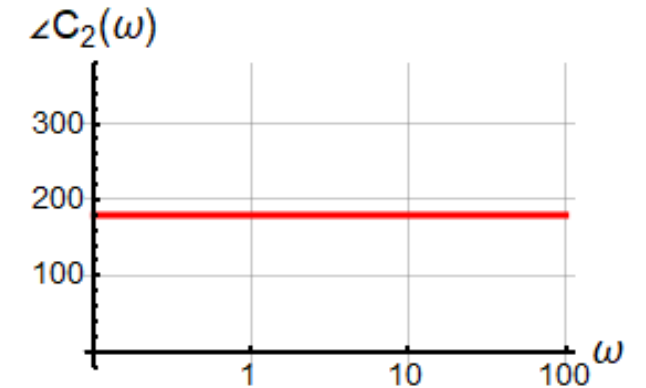
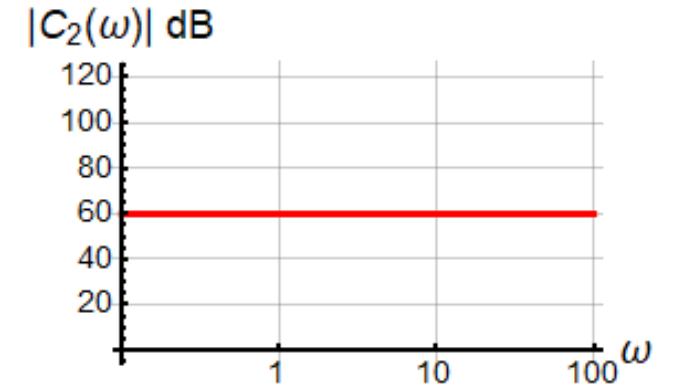
$$\angle C(\omega) = \angle \left(K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} \right) = \begin{cases} 0 & K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} \geq 0 \\ \pm 180^\circ & K \frac{b_1 b_2^{m_2} b_4^{m_3}}{a_1 a_2^{n_2} a_4^{n_3}} < 0 \end{cases}$$

A negative constant is represented as $|K| \angle 180^\circ$

$$C(\omega) = 1000$$



$$C(\omega) = -1000$$



Single $j\omega$ Factor

$$|\Omega(\omega)|_{dB} = m_1 \cdot 20 \cdot \log|\omega|$$

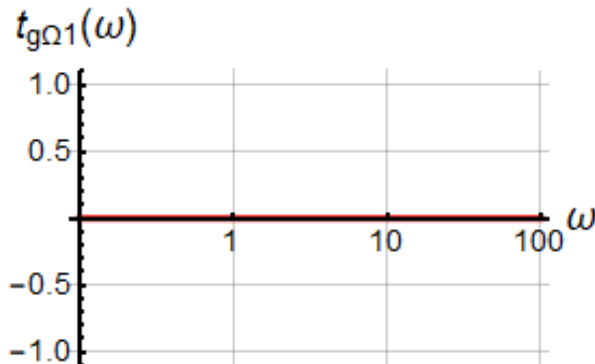
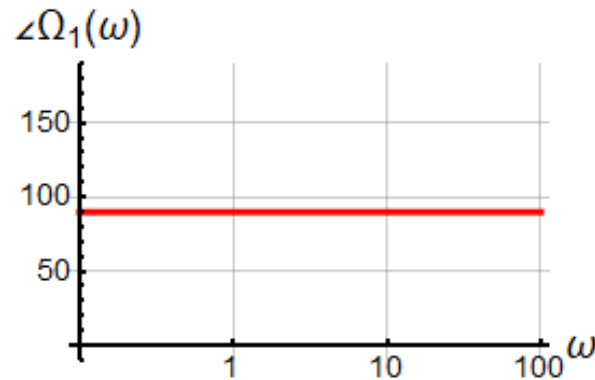
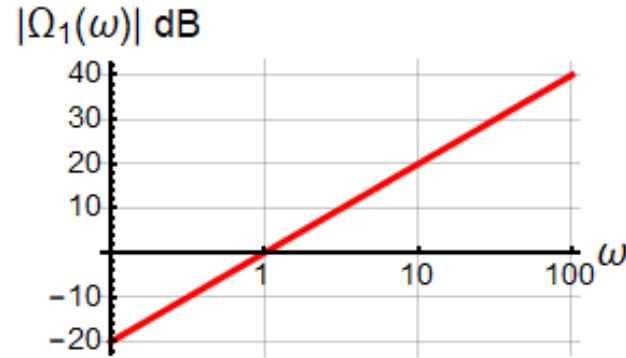
$$\angle\Omega(\omega) = m_1 \cdot \angle j\omega = m_1 \cdot 90^\circ$$

We are to plot the function $m_1 \times 20 \log_{10}|\omega|$ versus $\log_{10}|\omega|$.

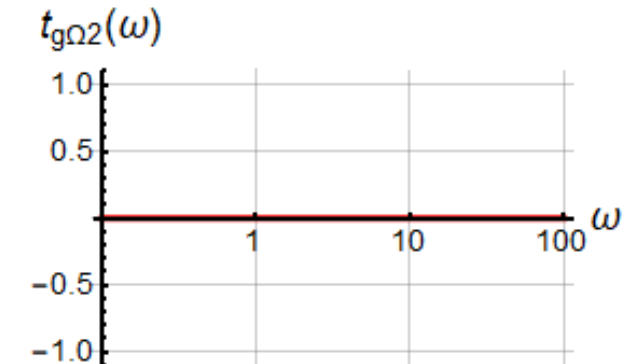
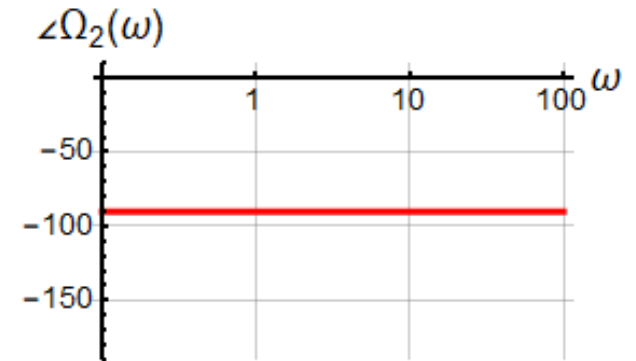
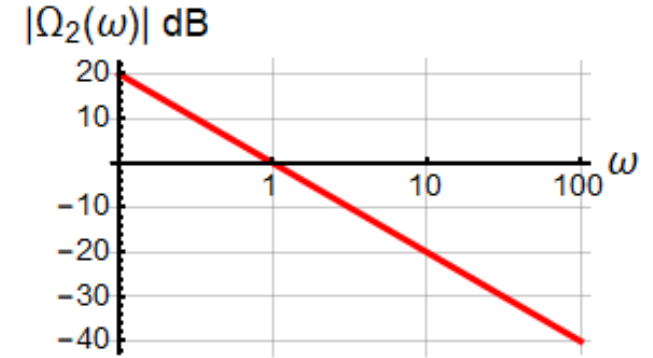
This gives a straight line with slope $\pm m_1 \times \frac{20dB}{decade}$.

The phase is simply $m_1 \times \angle \pm j = \pm m_1 \times 90^\circ$

$$m_1 = 1$$



$$m_1 = -1$$



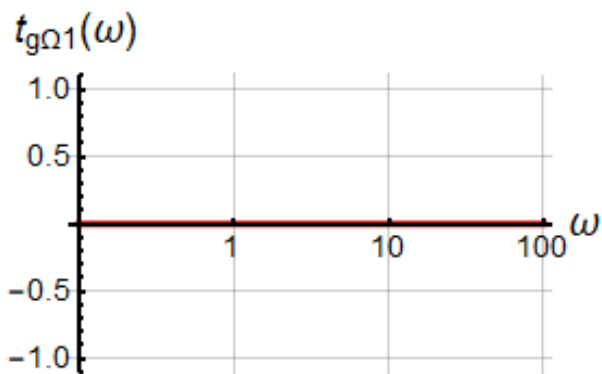
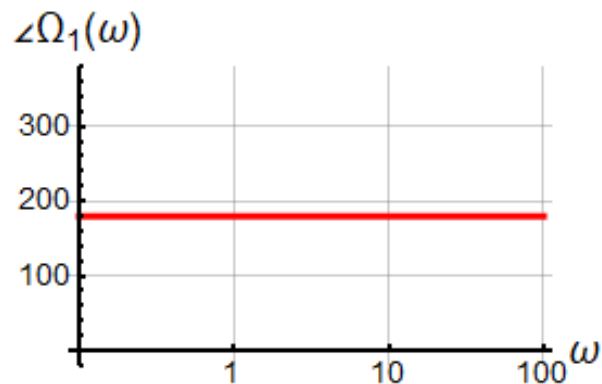
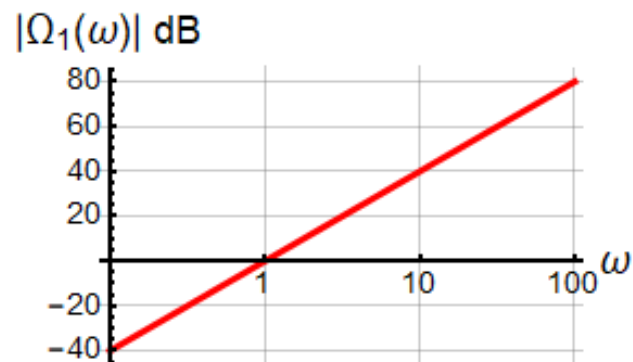
Double $j\omega$ Factor

$$|\Omega(\omega)|_{dB} = m_1 \cdot 20 \cdot \log|\omega|$$

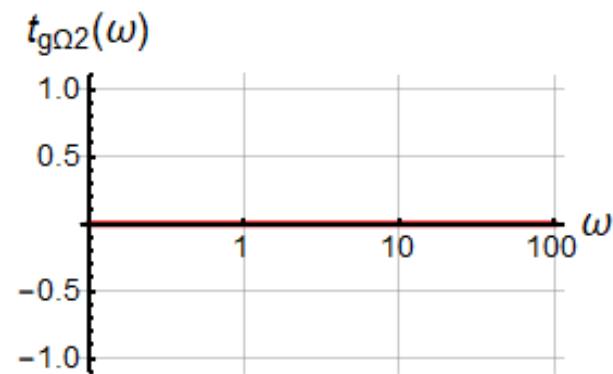
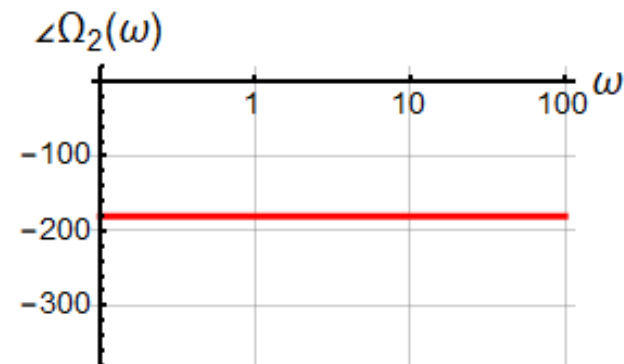
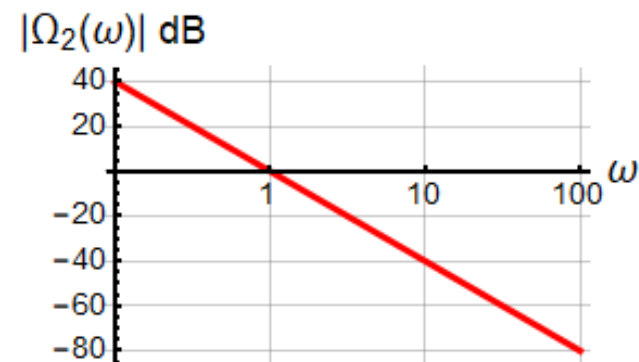
$$\angle\Omega(\omega) = m_1 \cdot \angle j\omega = m_1 \cdot 90^\circ$$

$$(j\omega)^{\pm 2}$$

$$m_1 = 2$$



$$m_1 = -2$$



We can write the amplitude of linear factors as:

$$\left(\frac{j\omega}{a_1} + 1\right)^{m_2} \Rightarrow |L(\omega)|_{dB} = m_2 \cdot 20 \cdot \log \left| \frac{j\omega}{a_1} + 1 \right|$$

To simplify the drawing of the linear function amplitude we consider the two cases:

Low frequency asymptote:

$$\omega \ll a_1 \quad |L(\omega)|_{dB} = m_2 \cdot 20 \cdot \log \left| \frac{j\omega}{a_1} + 1 \right| \approx m_2 \cdot 20 \cdot \log|1| = 0dB$$

High frequency asymptote:

$$\omega \gg a_1 \quad |L(\omega)|_{dB} = m_2 \cdot 20 \cdot \log \left| \frac{j\omega}{a_1} + 1 \right| \approx m_2 \cdot 20 \cdot \log \left| \frac{\omega}{a_1} \right|$$

Break frequency:

$$\omega_c = a_1 \quad |L(\omega_c)|_{dB} = m_2 \cdot 20 \log|j1 + 1| = m_2 \cdot 20 \log|\sqrt{2}| = m_2 \cdot 3dB$$

Low and high frequency asymptotes intersect at break frequency.

Again, we consider the two cases:

Low frequency asymptote:

High frequency asymptote:

Break frequency:

$$\left(\frac{j\omega}{a_1} + 1\right)^{m_2} \Rightarrow \angle L(\omega) = m_2 \cdot \angle \left(\frac{j\omega}{a_1} + 1\right) = m_2 \cdot \tan^{-1} \left(\frac{\omega}{a_1}\right)$$

$$\omega \ll a_1 \quad \angle L(\omega) = m_2 \cdot \angle \left(\frac{j\omega}{a_1} + 1\right) = m_2 \cdot \angle 1 = 0^\circ$$

$$\omega \gg a_1 \quad \angle L(\omega) = m_2 \cdot \angle \left(\frac{j\omega}{a_1} + 1\right) = m_2 \cdot \angle \left(\frac{j\omega}{a_1}\right) = m_2 \cdot 90^\circ$$

$$\omega_c = a_1 \quad \angle L(\omega_c) = m_2 \cdot \angle \left(\frac{j\omega_c}{a_1} + 1\right) = m_2 \cdot \angle(j1 + 1) = m_2 \cdot 45^\circ$$

The transition range is plotted as a straight line between:

$$\frac{\omega_c}{10} \leq \omega \leq 10 \cdot \omega_c$$

We can calculate the time delay of a frequency from the phase spectrum. It is the slope of the phase curve. The derivative is always a relative measure, as the constant reference disappears in the differentiation. It therefore gives the time delay of frequencies in one range relative to frequencies in another range. The frequencies in one range are thought of as a group of frequencies. Hence the delay is referred to as **group delay**.

A linear factor in the denominator causes a positive time delay. A linear factor in the numerator causes a negative time delay.

The largest time delay occurs at DC, where it is a multiple of the time constant.

No time delay occurs at infinite frequency, where all capacitors are short circuit, with only resistors left in the circuit.

$$H(\omega) = \frac{1}{(j\omega RC + 1)^{n_1}}$$

$$\angle H(\omega) = -n_1 \tan^{-1} \omega RC$$

$$t_g(\omega) = -n_1 \frac{d}{d\omega} \angle H(\omega) = -n_1 \frac{d}{d\omega} (-\tan^{-1} \omega RC)$$

$$t_g(\omega) = n_1 \frac{1}{(\omega RC)^2 + 1} \cdot \frac{d}{d\omega} (\omega RC)$$

$$t_g(\omega) = n_1 \frac{RC}{(\omega RC)^2 + 1}$$

$$t_g(0) = n_1 RC$$

$$t_g(\infty) = 0$$

$$|L(\omega)|_{dB} = m_2 \cdot 20 \cdot \log \left| \frac{j\omega}{a_1} + 1 \right|$$

$$\begin{aligned} \angle L(\omega) &= m_2 \cdot \angle \left(\frac{j\omega}{a_1} + 1 \right) \\ &= m_2 \cdot \tan^{-1} \left(\frac{\omega}{a_1} \right) \end{aligned}$$

$$a_1 = 5 = \frac{1}{\tau}$$

Amplitude
slope:

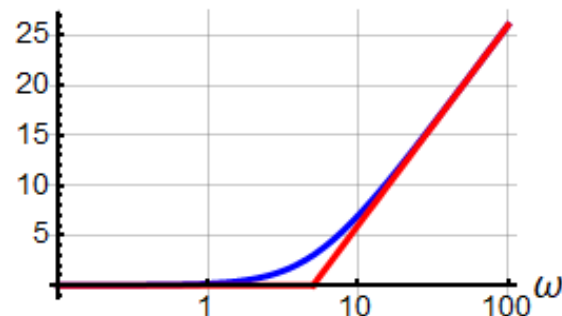
$$\begin{aligned} m_2 \cdot \frac{20dB}{decade} \\ m_2 \cdot \frac{6dB}{octave} \end{aligned}$$

Group
delay:

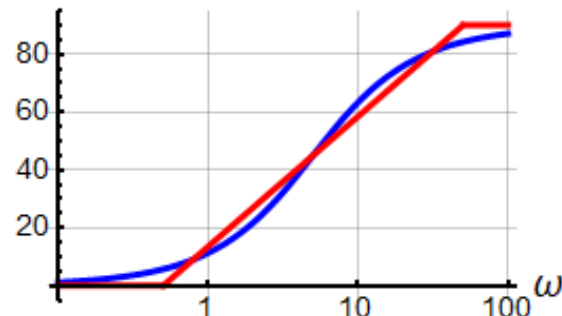
$$t_g = -m_2 \tau = -m_2 \times 0.2$$

$$m_2 = 1$$

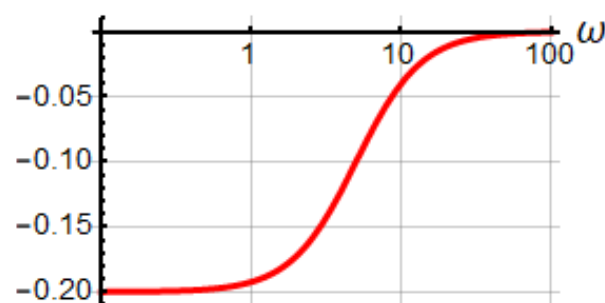
$|L_1(\omega)|$ dB



$\angle L_1(\omega)$

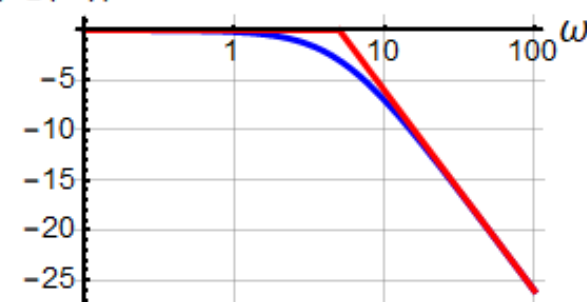


$t_{gL1}(\omega)$

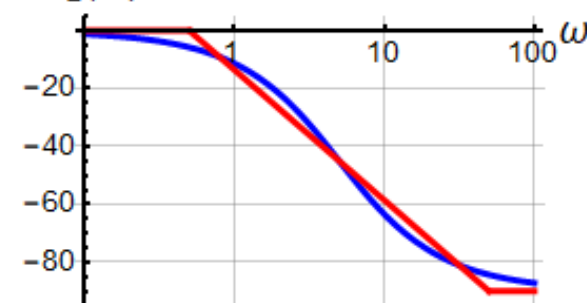


$$m_2 = -1$$

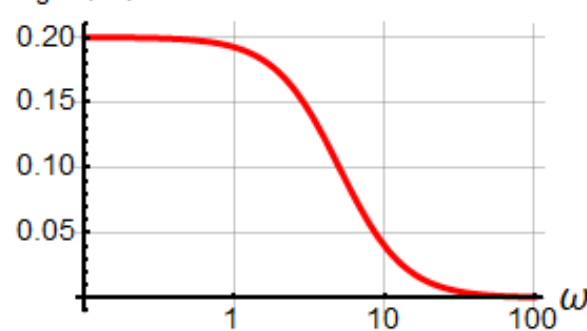
$|L_2(\omega)|$ dB



$\angle L_2(\omega)$



$t_{gL2}(\omega)$



$$|L(\omega)|_{dB} = m_2 \cdot 20 \log \left| \frac{j\omega}{a_1} + 1 \right|$$

$$\begin{aligned} \angle L(\omega) &= m_2 \cdot \angle \left(\frac{j\omega}{a_1} + 1 \right) \\ &= m_2 \cdot \tan^{-1} \left(\frac{\omega}{a_1} \right) \end{aligned}$$

$$a_1 = 5 = \frac{1}{\tau}$$

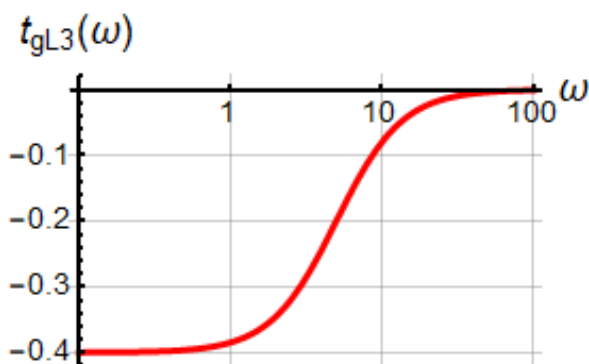
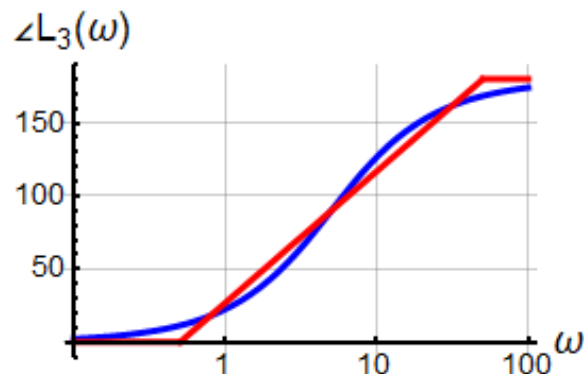
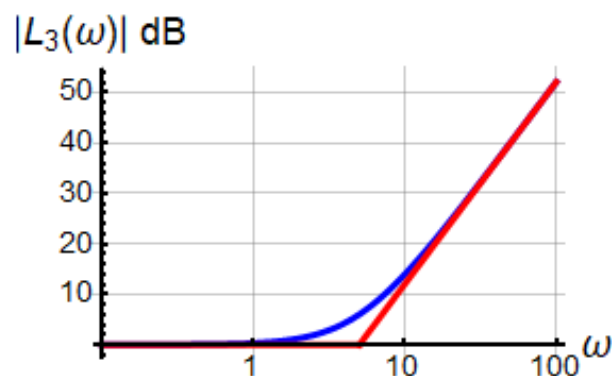
Amplitude
slope:

$$\begin{aligned} m_2 \cdot \frac{20dB}{decade} \\ m_2 \cdot \frac{6dB}{octave} \end{aligned}$$

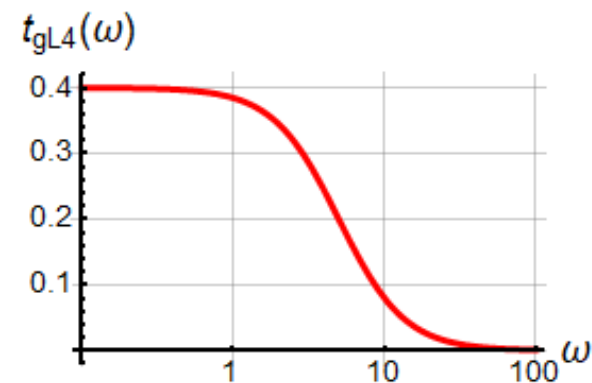
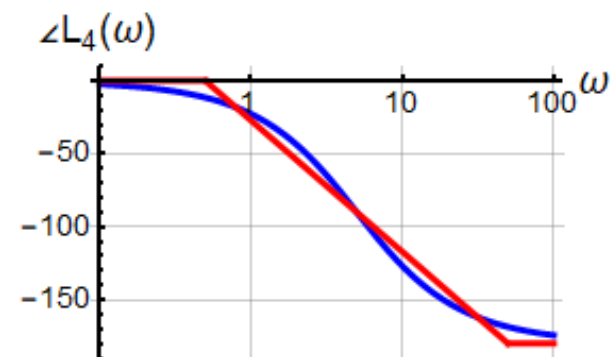
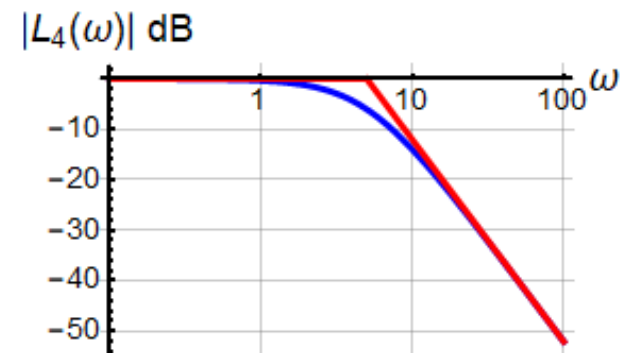
Group
delay:

$$t_g = -m_2 \tau = -m_2 \times 0.2$$

$$m_2 = 2$$



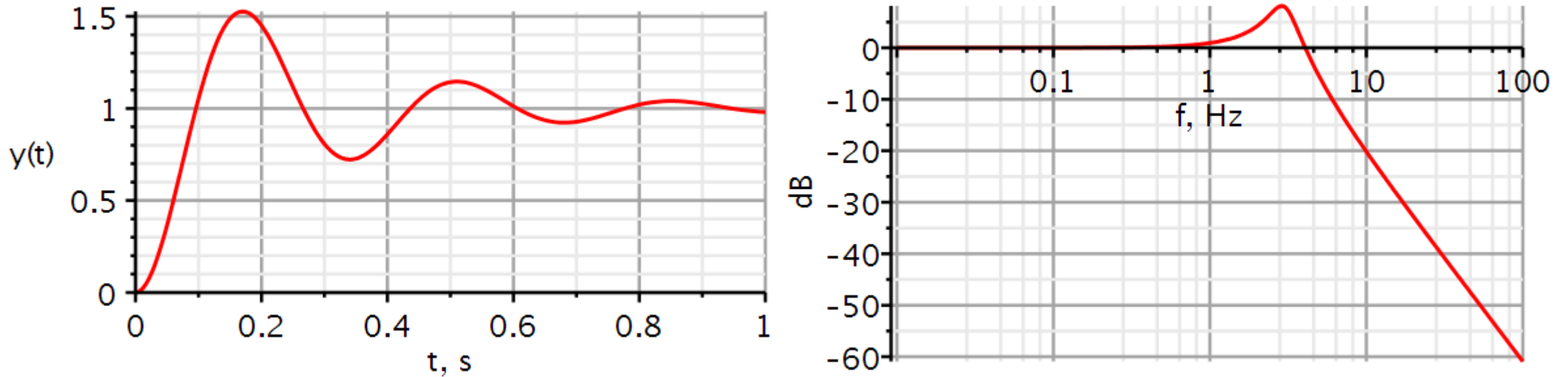
$$m_2 = -2$$



Problems

Problem 1: Systems identification from step response

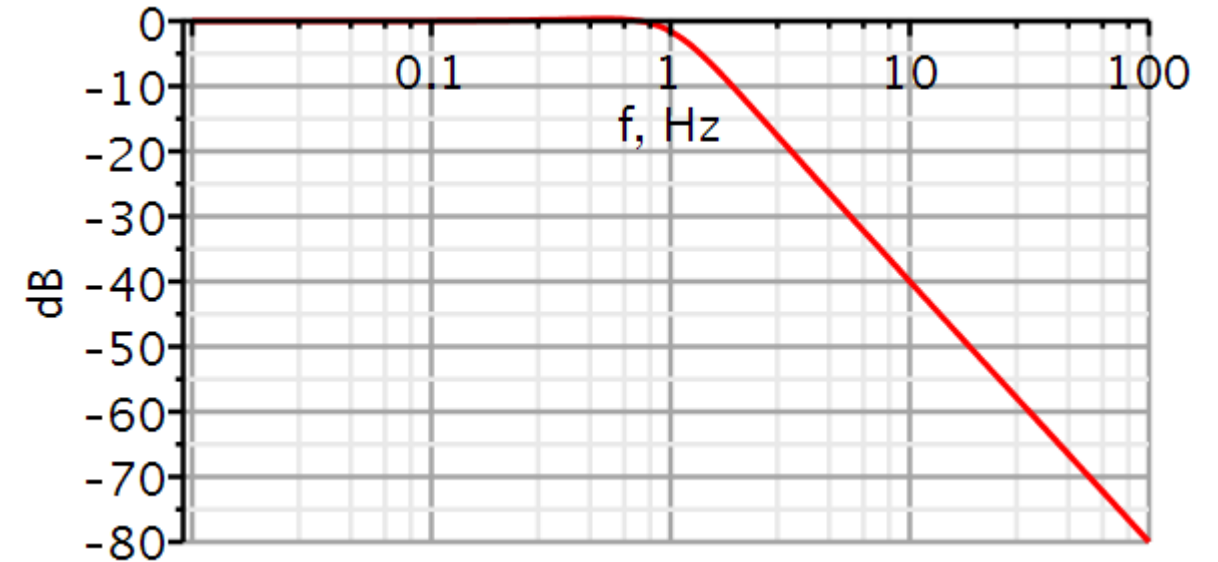
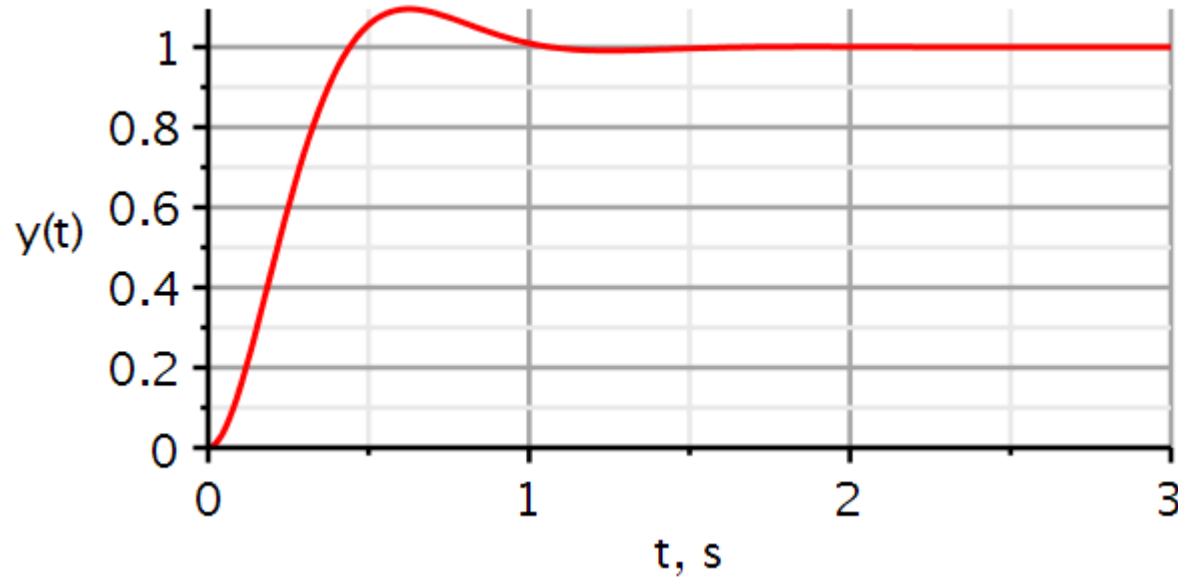
From an unknown system the following step response and amplitude spectrum were measured.



Identify the system parameters, write down the differential equation and plot the amplitude response using the identified systems parameters.

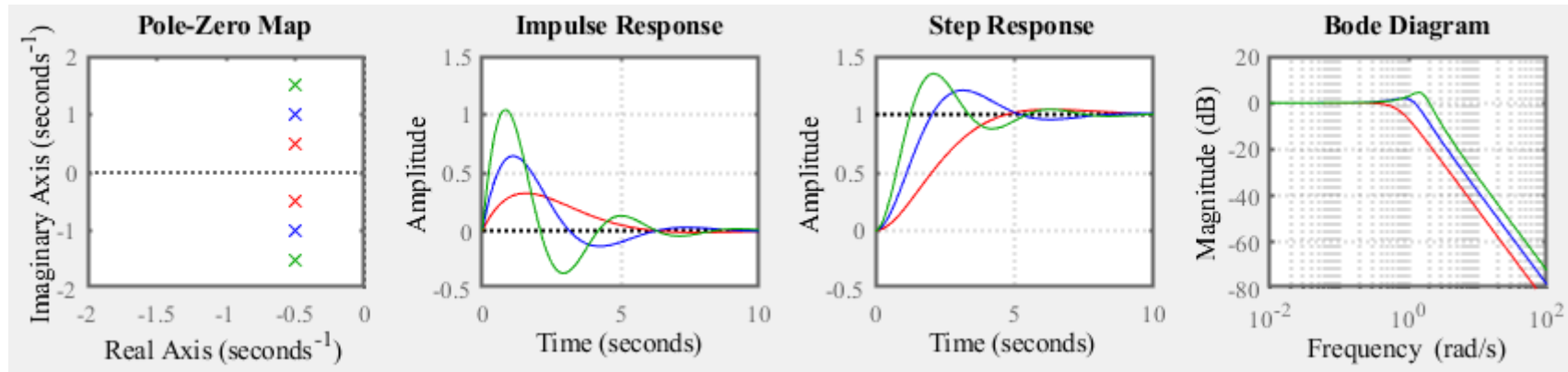
Problem 2: Systems identification from step response

From an unknown system the following step response and amplitude spectrum were measured.

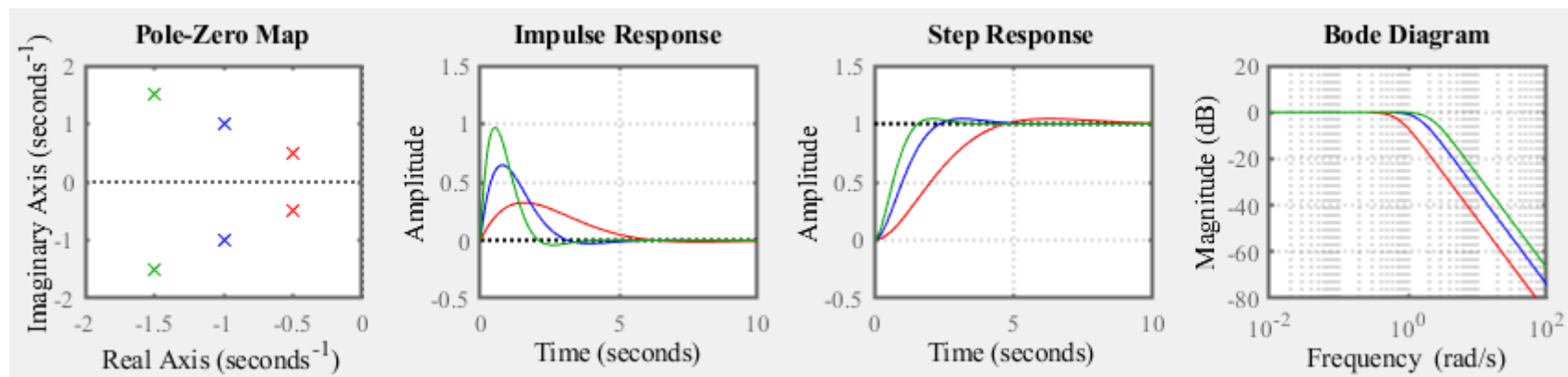


Identify the system parameters, write down the differential equation and plot the amplitude response using the identified systems parameters.

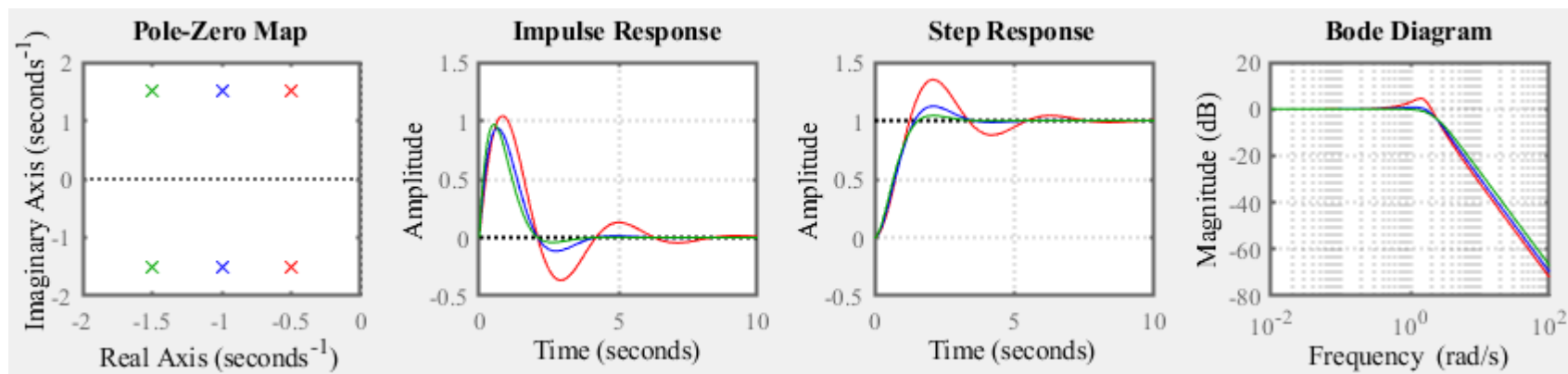
Problem 3: Explain the effect of poles moving



Problem 3: Explain the effect of poles moving



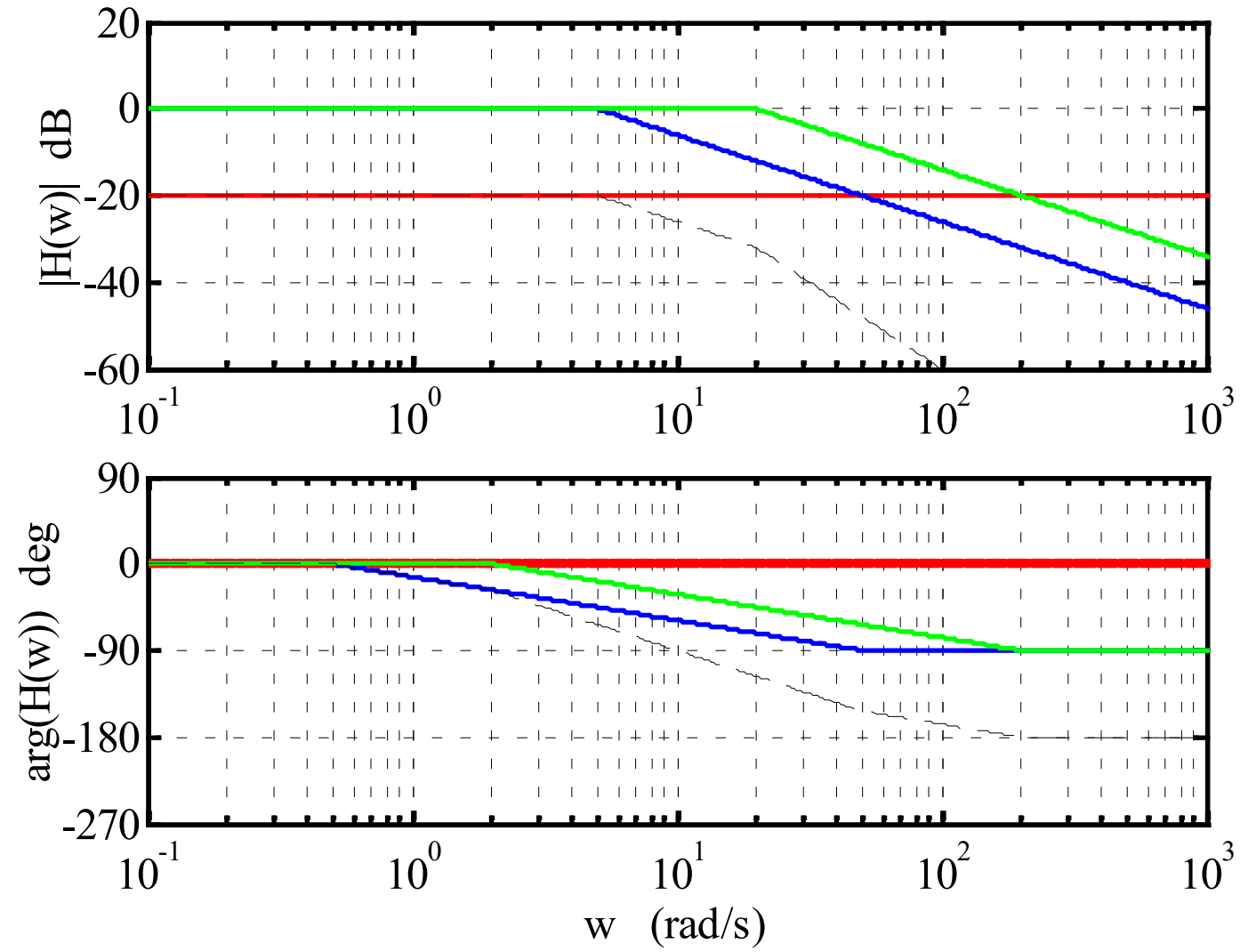
Problem 3: Explain the effect of poles moving



Problem 4: Bode plot

Analyze the Bode plot and identify the transfer function.

Also predict the group delay t_g at $\omega = 0$ for each linear factor.



Problem 4: Bode plot – redraw using Maple

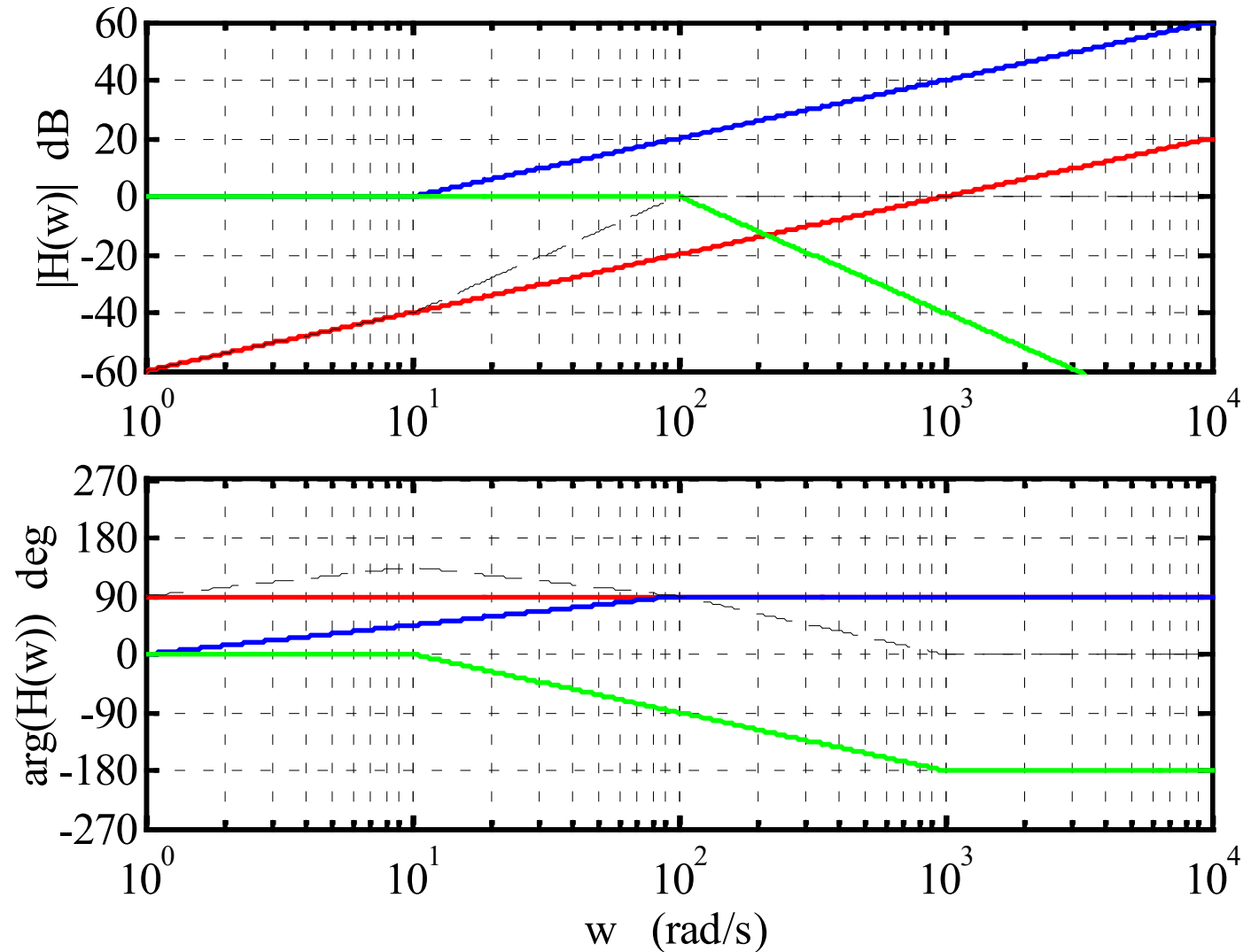
Download Maple file from DTU Learn.

Modify the script to plot the Bode plot curves of the present system.

Problem 5: Bode plot

Analyze the Bode plot and identify the transfer function.

Also predict the group delay t_g at $\omega = 0$ for each linear factor.



Problem 5: Bode plot – redraw using Maple

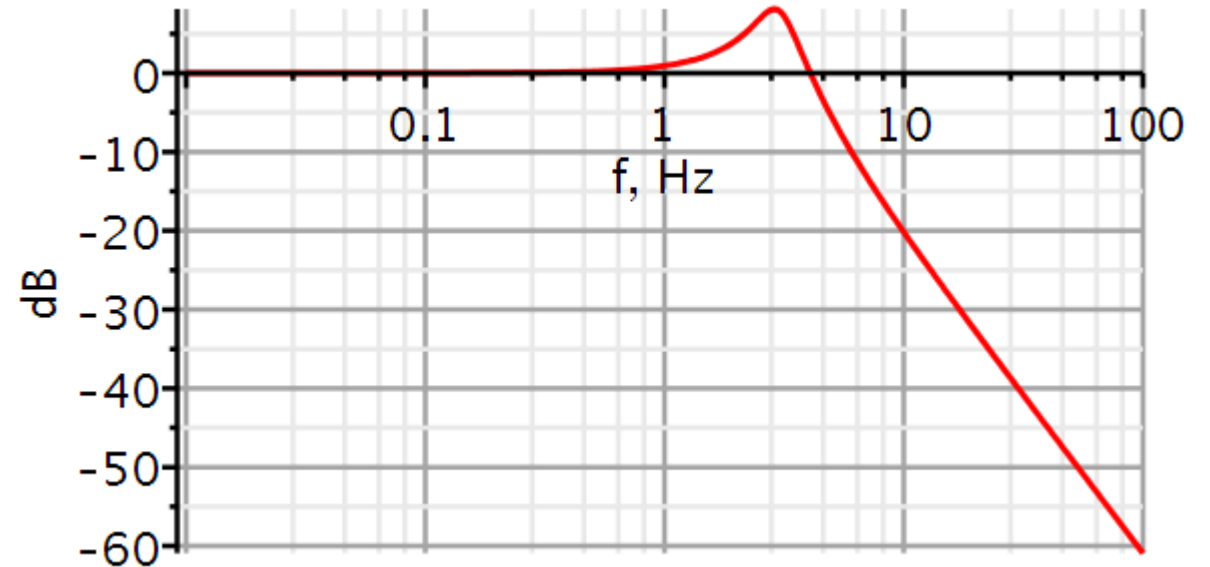
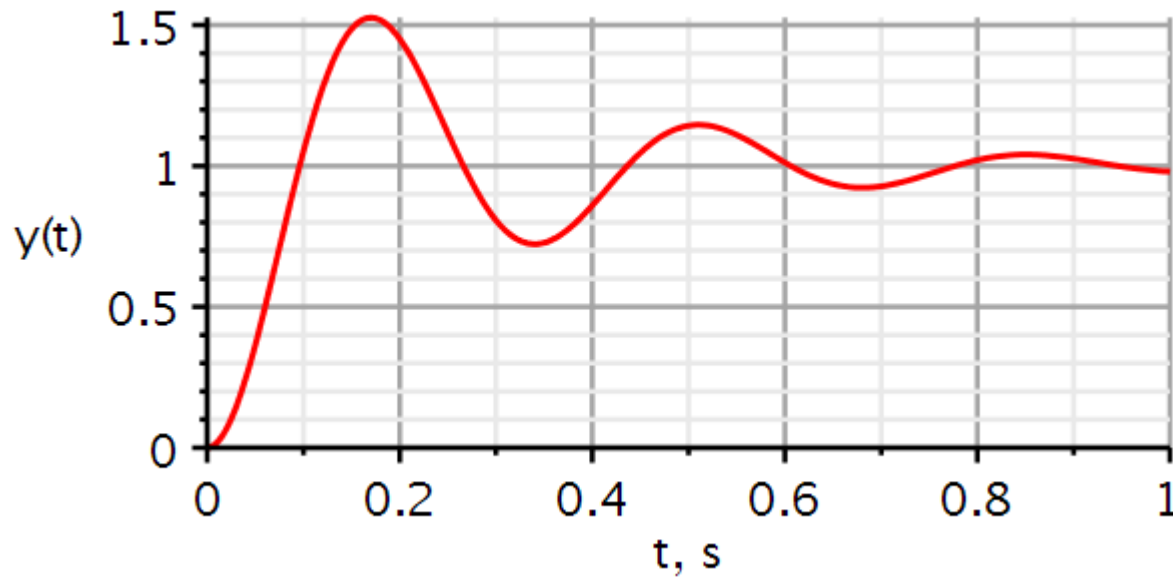
Download Maple file from DTU Learn.

Modify the script to plot the Bode plot curves of the present system.

Solutions

Problem 1: Systems identification from step response

From an unknown system the following step response and amplitude spectrum were measured.



Identify the system parameters, write down the differential equation and plot the amplitude response using the identified systems parameters.

Problem 1: Systems identification from step response (sol)

$$PO = 52$$

$$a \stackrel{\text{def}}{=} \ln\left(\frac{PO}{100}\right) = \ln 0.52 = -0.654 \quad \rightarrow$$

$$\zeta = \frac{|a|}{\sqrt{\pi^2 + a^2}} = \frac{0.654}{\sqrt{\pi^2 + 0.654^2}} = 0.204$$

$$t_p = 0.168$$

$$\omega_n = \frac{\pi}{t_p \sqrt{1 - \zeta^2}} = \frac{\pi}{0.168 \sqrt{1 - 0.204^2}} = 19.1$$

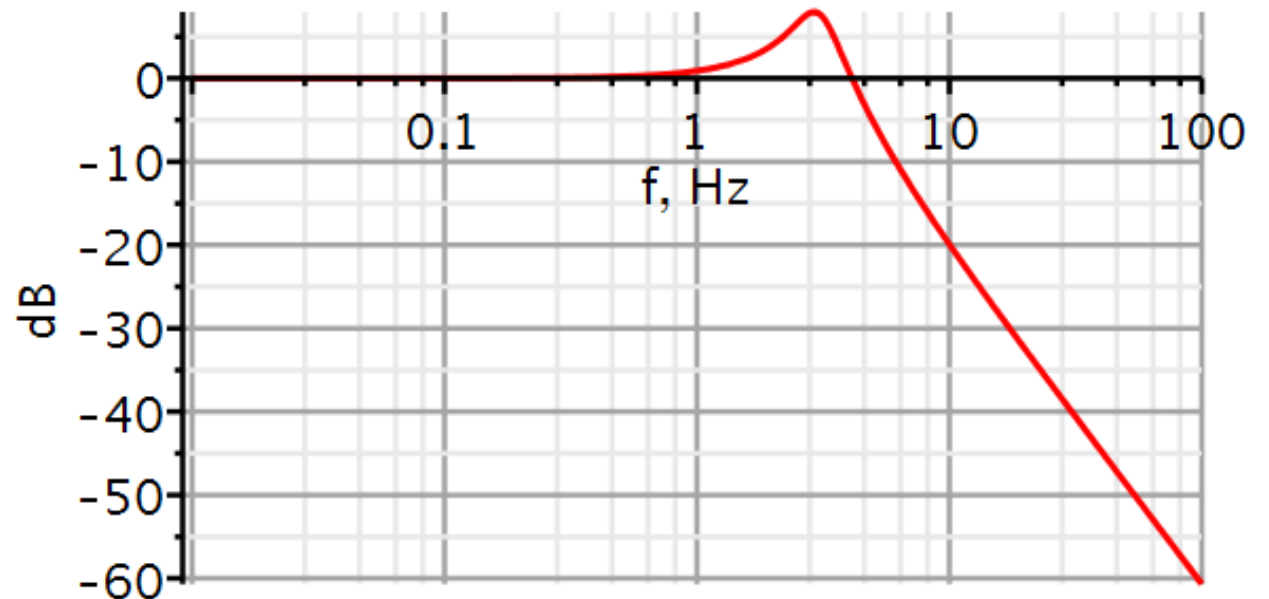
$$a_1 = 2\zeta\omega_n = 7.79 \quad a_0 = \omega_n^2 = 364.7$$

$$\ddot{y}(t) + 7.79\dot{y}(t) + 364.7y(t) = 364.7x(t)$$

Identified systems

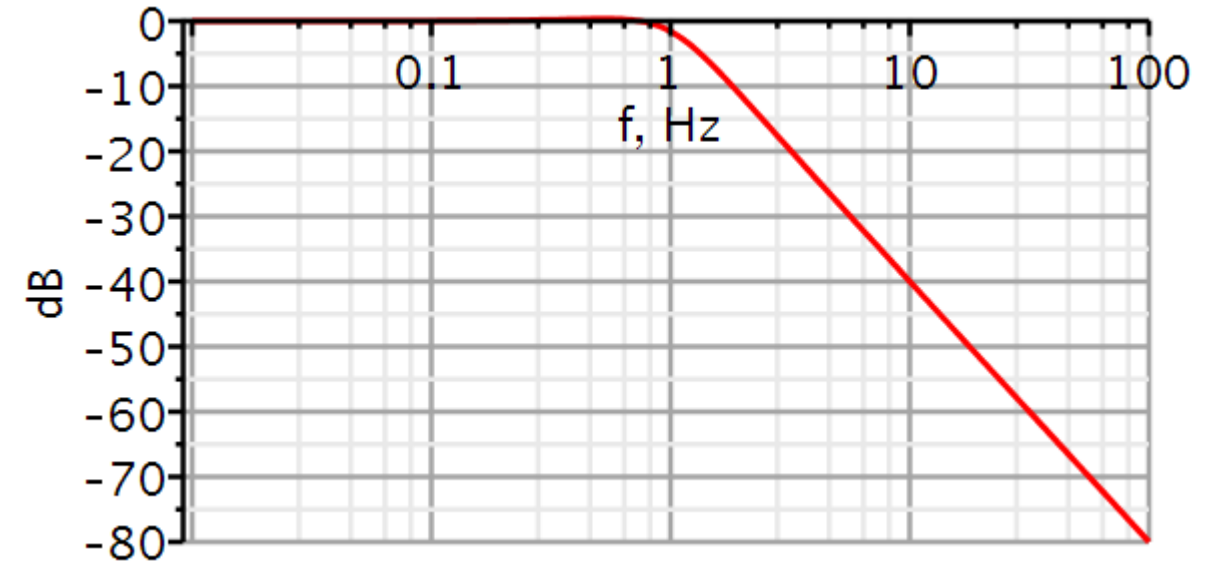
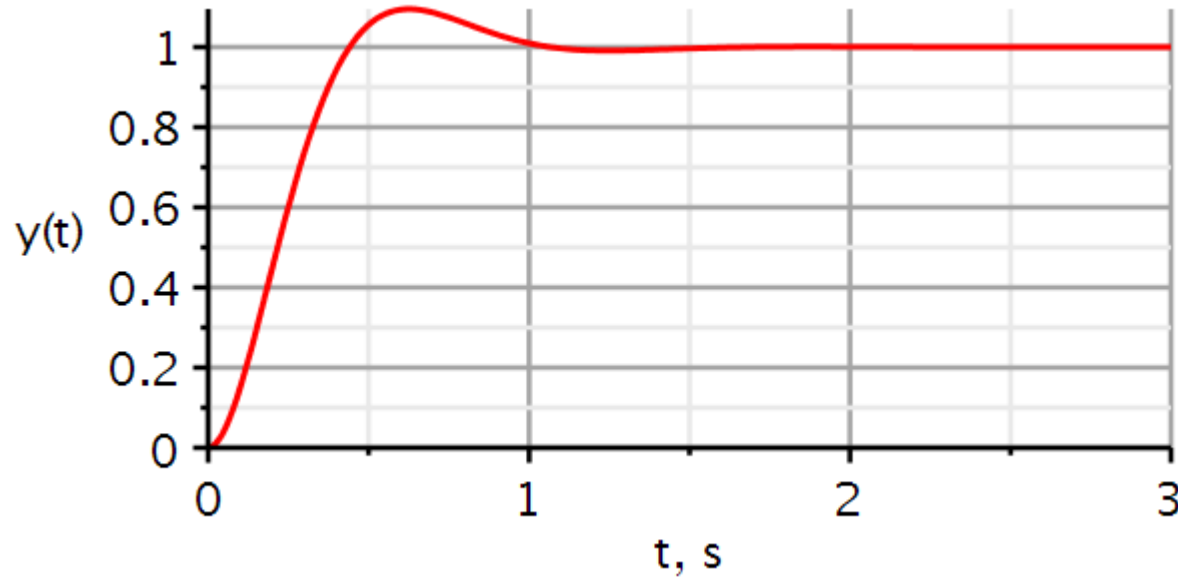
$$\text{sys11est} := \text{Coefficients}\left(\frac{(19.1)^2}{s^2 + 2 \cdot 0.204 \cdot 19.1 \cdot s + (19.1)^2}\right):$$

`MagnitudePlot(sys11est, color = [red, blue], thickness = 3, axesfont = ["Helvetica", "ROMAN", 18], axis[2] = [thickness = 2.5], axis[1] = [mode = log, thickness = 2.5], labels = ["f, Hz", "dB"], labelfont = ["HELVETICA", 18], gridlines, decibels = true, hertz = true, size = [600, 300])`



Problem 2: Systems identification from step response

From an unknown system the following step response and amplitude spectrum were measured.



Identify the system parameters, write down the differential equation and plot the amplitude response using the identified systems parameters.

Problem 2: Systems identification from step response (sol)

$$PO = 9.5$$

$$a \stackrel{\text{def}}{=} \ln\left(\frac{PO}{100}\right) = \ln 0.095 = -2.354 \quad \rightarrow$$

$$\zeta = \frac{|a|}{\sqrt{\pi^2 + a^2}} = \frac{2.354}{\sqrt{\pi^2 + 2.354^2}} = 0.599$$

$$t_p = 0.615$$

$$\omega_n = \frac{\pi}{t_p \sqrt{1 - \zeta^2}} = \frac{\pi}{0.615 \sqrt{1 - 0.599^2}} = 6.379$$

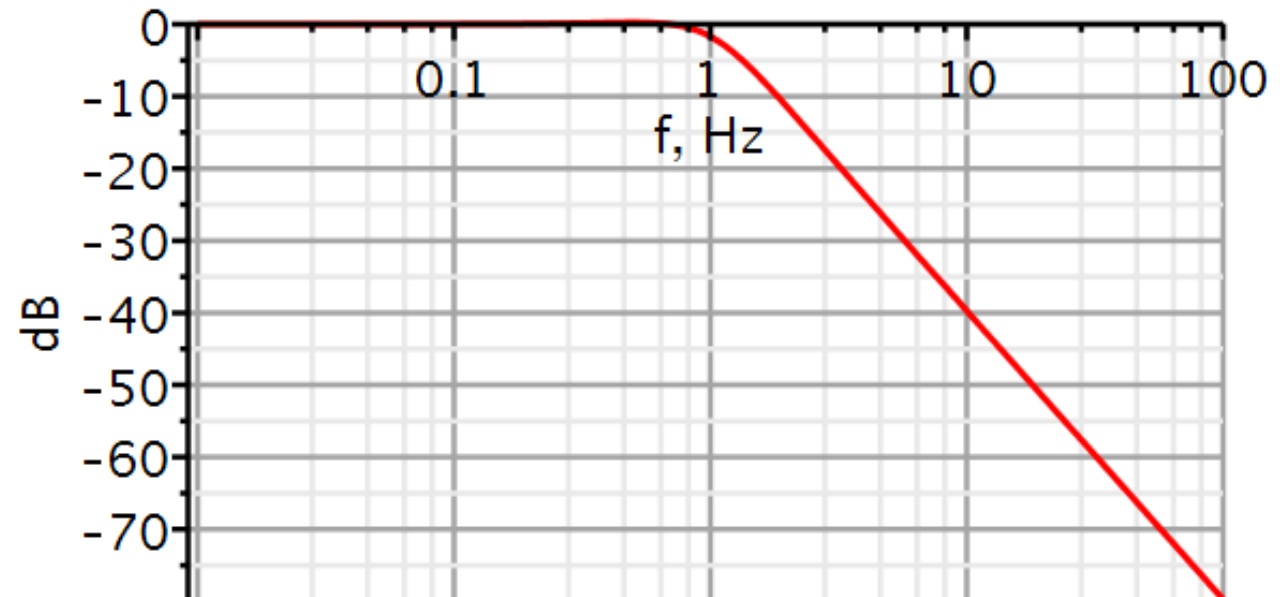
$$a_1 = 2\zeta\omega_n = 7.64 \quad a_0 = \omega_n^2 = 40.69$$

$$\ddot{y}(t) + 7.64\dot{y}(t) + 40.69y(t) = 40.69x(t)$$

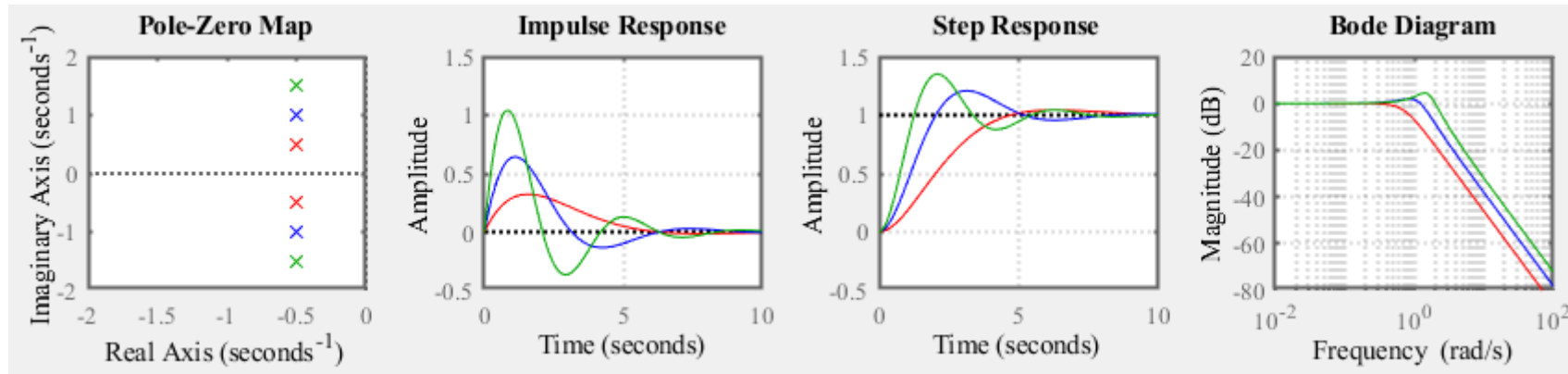
Identified systems

$$\text{sys11est} := \text{Coefficients}\left(\frac{(6.379)^2}{s^2 + 2 \cdot 0.599 \cdot 6.379 \cdot s + (6.379)^2}\right):$$

MagnitudePlot(sys11est, color = [red, blue], thickness = 3, axesfont = ["Helvetica", "ROMAN", 18], axis[2] = [thickness = 2.5], axis[1] = [mode = log, thickness = 2.5], labels = ["f, Hz", "dB "], labelfont = ["HELVETICA", 18], gridlines, decibels = true, hertz = true, size = [600, 300])



Problem 3: Explain the effect of poles moving (sol)



As the poles move downward from green to red, the angle with the negative real axis becomes smaller and the damping ratio ζ increases. At the same time, the undamped natural frequency ω_n is reduced and the time to peak t_p is increased.

This agrees with the decreasing amplitudes of the oscillations in the impulse and step responses. Also, the frequency of the magnitude peak is decreased.

$$PO = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

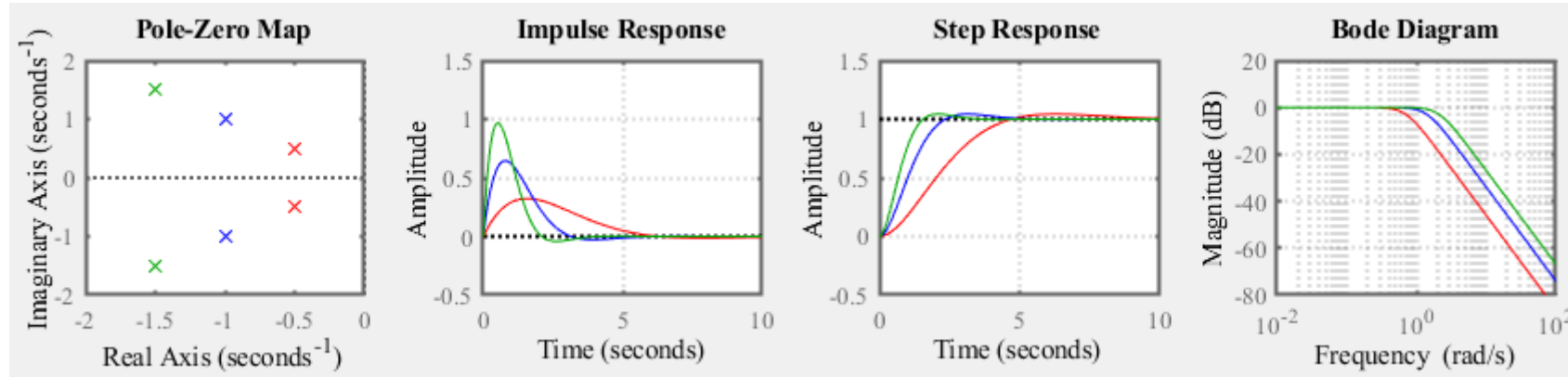
$$\text{If } 0 < \zeta < 1/\sqrt{2}: \quad \omega_{peak} = \omega_n\sqrt{1-2\zeta^2}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$M_{pk} = m_3 \cdot 20 \cdot \log \left[2\zeta\sqrt{1-\zeta^2} \right] \text{ dB}$$

Problem 3: Explain the effect of poles moving (sol)



Here the natural frequency is reduced when the poles move from green to red. On the other hand, the angle remains unchanged, hence the damping factor is unchanged.

We see that the overshoot in the step response is unchanged, but the time to peak is increased.

There are no peaks in the amplitude spectrum, but the cut-off frequency is reduced because the natural frequency is reduced.

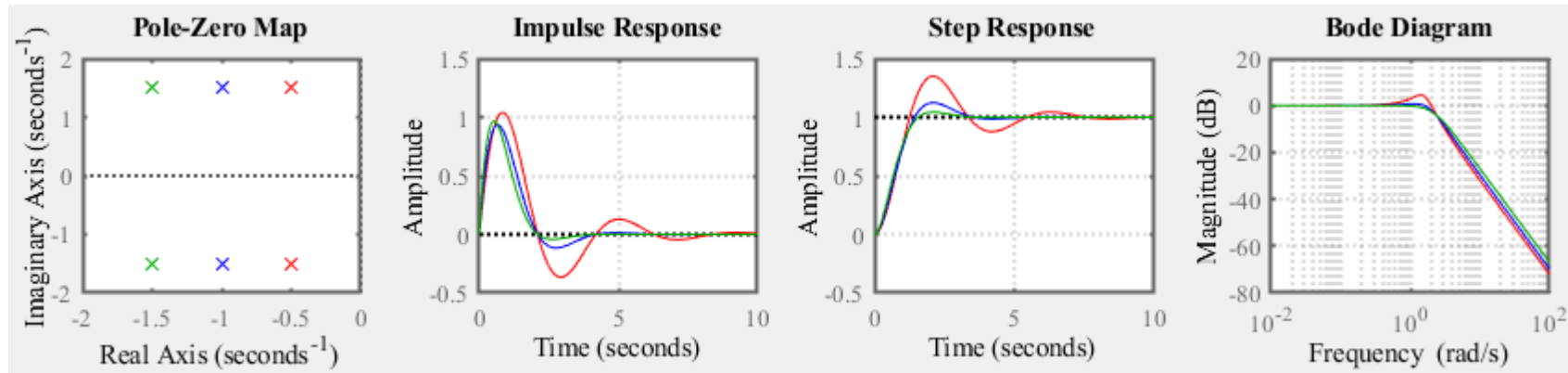
$$PO = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$\text{If } 0 < \zeta < 1/\sqrt{2}: \quad \omega_{peak} = \omega_n\sqrt{1-2\zeta^2}$$

$$M_{pk} = m_3 \cdot 20 \cdot \log \left[2\zeta\sqrt{1-\zeta^2} \right] \text{ dB}$$

Problem 3: Explain the effect of poles moving (sol)



Here the damped frequency is constant when the poles move from green to red. On the other hand, the angle increases, hence the damping factor is reduced.

We see that the overshoot in the step response increases, but the time to peak is unchanged.

There appears a peak in the amplitude spectrum when the system becomes more underdamped, but the cut-off frequency is *almost* unchanged.

$$PO = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

$$\text{If } 0 < \zeta < 1/\sqrt{2}: \quad \omega_{peak} = \omega_n\sqrt{1-2\zeta^2}$$

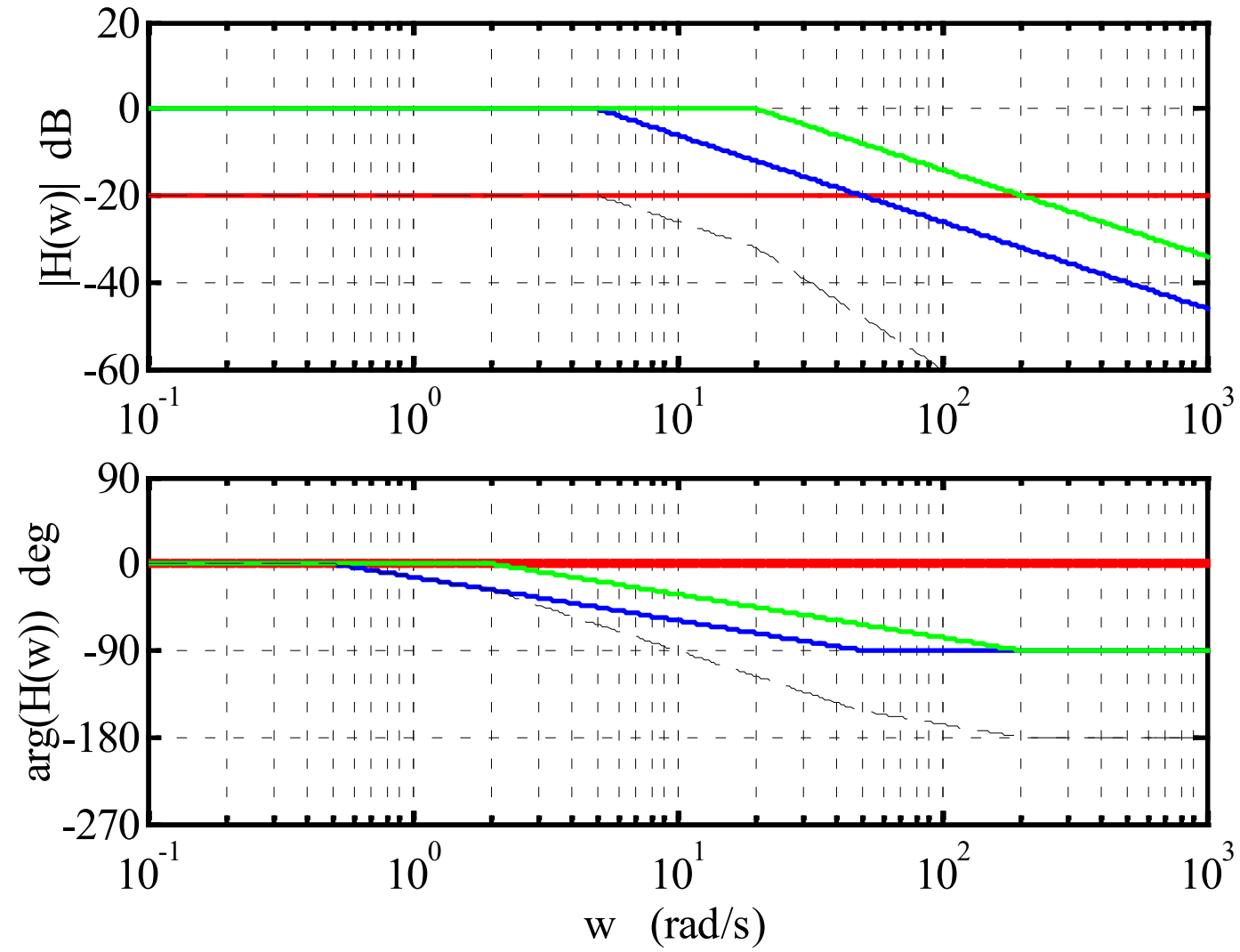
$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$M_{pk} = m_3 \cdot 20 \cdot \log \left[2\zeta\sqrt{1-\zeta^2} \right] \text{ dB}$$

Problem 4: Bode plot

Analyze the Bode plot and identify the transfer function.

Also predict the group delay t_g at $\omega = 0$ for each linear factor.



Problem 4: Bode plot

Red curve:

$$C(\omega) = -20 \text{ dB} = 0.1$$

$$t_g = -n_1\tau$$

Blue curve:

$$L_1(\omega) = \left(\frac{j\omega}{5} + 1\right)^{-1}$$

$$\tau_1 = 0.2\text{s}$$

$$t_{g_1} = -(-1) \times 0.2\text{s} = 0.2\text{s}$$

Green curve:

$$L_2(\omega) = \left(\frac{j\omega}{20} + 1\right)^{-1}$$

$$\tau_2 = 0.05\text{s}$$

$$t_{g_2} = -(-1) \times 0.05\text{s} = 0.05\text{s}$$

$$H(\omega) = \frac{0.1}{\left(\frac{j\omega}{5} + 1\right)\left(\frac{j\omega}{20} + 1\right)} = \frac{10}{(j\omega + 5)(j\omega + 20)}$$

$$H(s) = \frac{10}{(s + 5)(s + 20)} = \frac{10}{s^2 + 25s + 100}$$

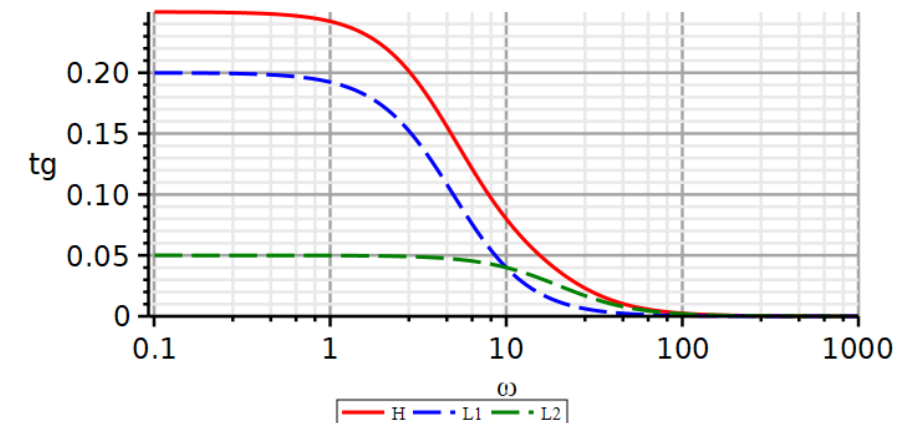
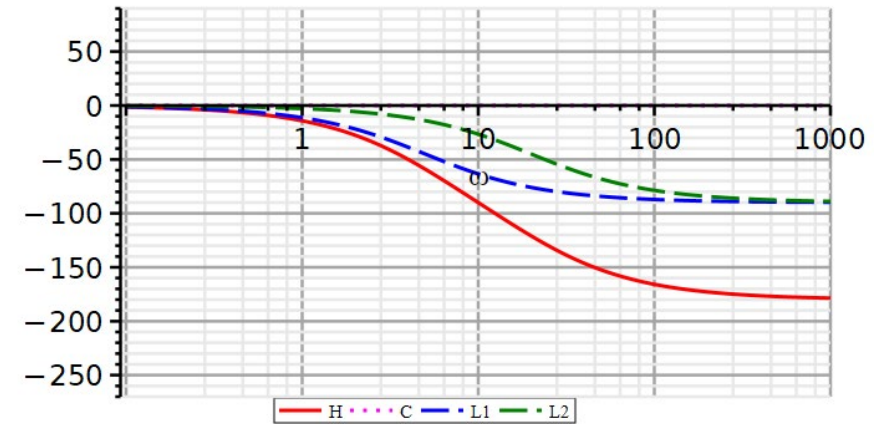
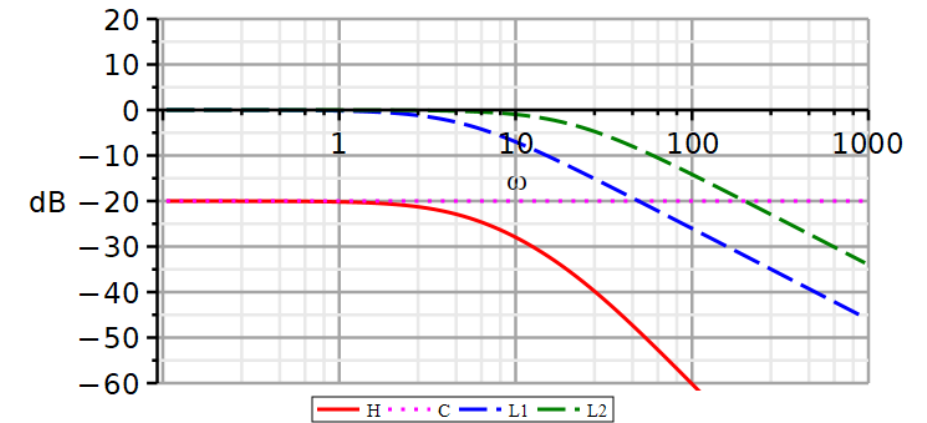
Problem 4: Bode plot – redraw using Maple

Download Maple file from DTU Learn.

Modify the script to plot the Bode plot curves of the present system.

The system phase curve is the sum of the phase curves of each individual factor.

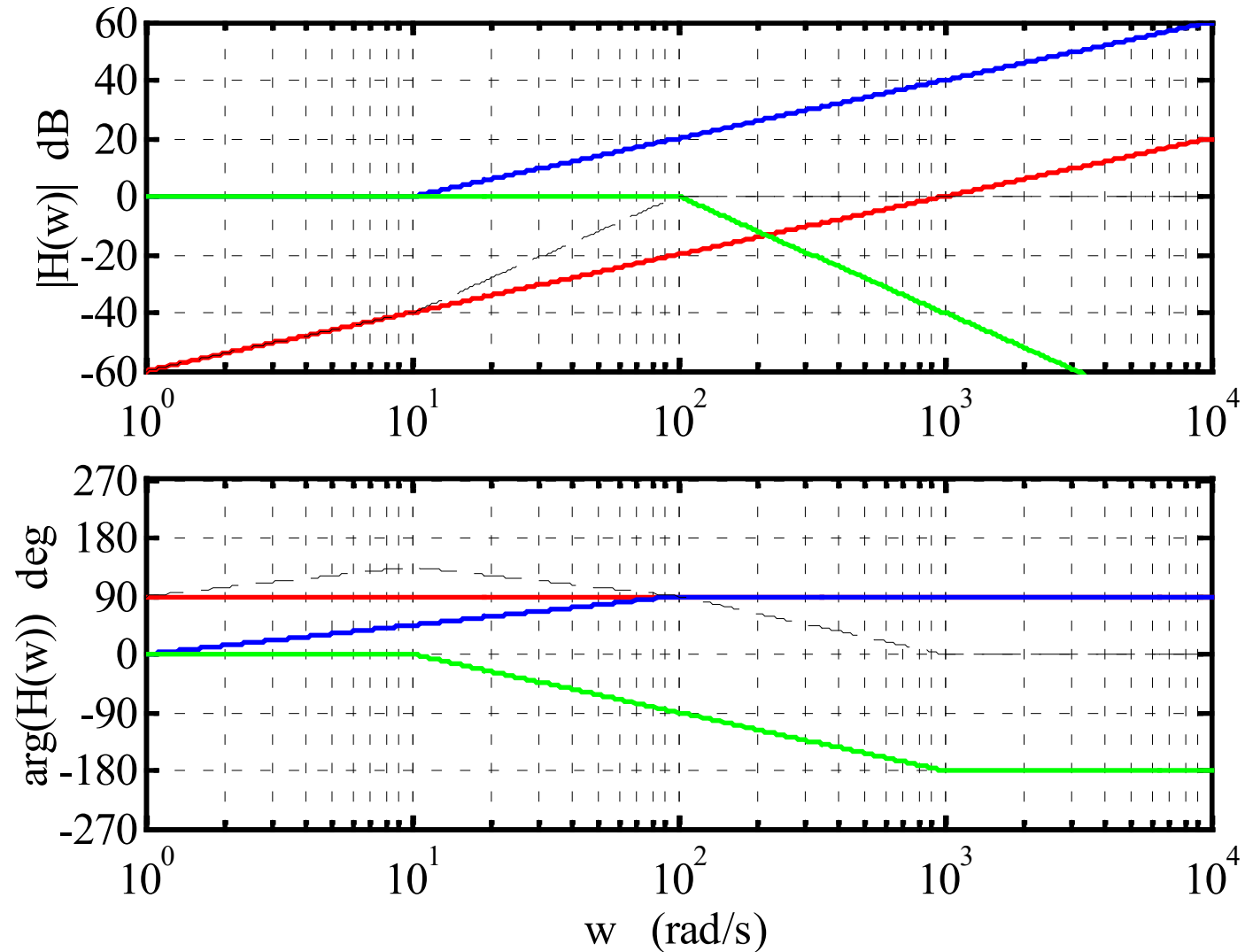
Because the phase curves add together, so does the group delays caused by the individual factors.



Problem 5: Bode plot

Analyze the Bode plot and identify the transfer function.

Also predict the group delay t_g at $\omega = 0$ for each linear factor.



Problem 5: Bode plot

Red curve:

$$\Omega(\omega) = \frac{j\omega}{10^3}$$

$$t_g = -n_1\tau$$

Blue curve:

$$L_1(\omega) = \left(\frac{j\omega}{10} + 1\right) \quad \tau_1 = 0.1s$$

$$t_{g_1} = -1 \times 0.1s = -0.1s$$

Green curve:

$$L_2(\omega) = \left(\frac{j\omega}{100} + 1\right)^{-2} \quad \tau_2 = 0.01s$$

$$t_{g_2} = -(-2) \times 0.01s = 0.02s$$

$$H(\omega) = \frac{\frac{j\omega}{10^3} \left(\frac{j\omega}{10} + 1\right)}{\left(\frac{j\omega}{100} + 1\right)^2} = \frac{j\omega(j\omega + 10)}{(j\omega + 100)^2}$$

$$H(s) = \frac{s(s + 10)}{(s + 100)^2}$$

Problem 5: Bode plot – redraw using Maple

Download Maple file from DTU Learn.

Modify the script to plot the Bode plot curves of the present system.

The system phase curve is the sum of the phase curves of each individual factor.

Because the phase curves add together, so does the group delays caused by the individual factors.

